Dispatching Problem Definition

1. Unified Problem Classification

The comprehensive fulfillment optimization problem is formally classified as the **Extended Rich Multi-Depot, Multi-Trip, Multi-Commodity Pickup-and-Delivery Problem with Open Time Windows, 3D Containerization, Human-Robot Shared Spaces, and Stochastic Disruption Recourse** ($\mathcal{P}_{\text{ER-MD-VRPTW-3D-HRI}}$).

Below is the extended mathematical architecture, integrating the foundational operational structure with six additional real-world warehouse restrictions.

Extended Notation, Sets, and Parameters

Extended Graph and Topological Sets

- $\mathcal{V} = \mathcal{V}_0 \cup \mathcal{V}_P \cup \mathcal{V}_D \cup \mathcal{V}_{\text{ASRS}} \cup \mathcal{V}_{\text{charge}}$: Extended vertex set.
  - $\mathcal{V}_{\text{ASRS}} \subset \mathcal{V}_P$: Automated Storage and Retrieval System (AS/RS) induction spurs, vertical lift modules (VLMs), and mini-load crane pick faces.
- $\mathcal{Z} = \mathcal{Z}_{\text{auto}} \cup \mathcal{Z}_{\text{mixed}} \cup \mathcal{Z}_{\text{human}}$: Partition of physical floor zones into fully automated AMR zones, mixed cobot/human aisles, and pedestrian-only manual picking galleries.
- $\mathcal{I}$: Discrete set of physical cargo items/SKUs, where each item $i \in \mathcal{I}$ has a 3D bounding box dimension $\mathbf{dim}_i = [l_i, w_i, h_i]^\top$, mass m_i, fragility rating $\rho_i \in (0, 1]$, and maximum allowable compressive load $\sigma_i^{\max}$.
- $\mathcal{B}_k$: Array of discrete onboard physical tote slots or roll-cage bays in vehicle $k \in \mathcal{K}$.

Additional Operational Parameters

- $\mathbf{v}_{\text{safe}}(\mathcal{Z})$: Maximum permissible velocity vector inside zone $\mathcal{Z}$, where $\mathbf{v}_{\text{safe}}(\mathcal{Z}_{\text{mixed}}) \ll \mathbf{v}_{\text{safe}}(\mathcal{Z}_{\text{auto}})$ pursuant to ISO 3691-4 industrial safety standards.
- $R_k^{\text{sweep}}(\theta)$: Dynamic polygonal footprint/swept swept-volume radius of vehicle k during rotational maneuvers at angular displacement θ.
- μ_c : Continuous processing/packing capacity rate of consolidation station $c \in \mathcal{V}_D$ (items per unit time).
- $t_{\text{crane}}(\ell_1, \ell_2)$: Deterministic or scheduled transit duration of the AS/RS crane carriage between vertical coordinates $\ell_1, \ell_2 \in \mathbb{R}^3$.
- $p_i^{\text{stockout}} \in [0, 1)$: Empirical probability of an inventory

discrepancy/phantom stock event at pick face $i \in \mathcal{V}_P$.

- $1-\epsilon$: Required probabilistic confidence level for on-time order delivery (chance constraint).

2. Constraints

2.1. Spatial Topology and Routing Constraints

- **Directed Graph Traversability:** Routing is restricted to feasible directed arcs $(i, j) \in \mathcal{A}$; transitions across unlinked coordinates (e.g., across physical rack uprights) are strictly forbidden:
  $$x_{ij}^k = 0 \quad \forall (i,j) \notin \mathcal{A}$$
- **One-Way Aisle Flow:** Aisle directionality mandates unidirectional movement along specific graph subsets:
  $$x_{ji}^k = 0 \quad \forall (i,j) \in \mathcal{A}_{\text{one-way}},\ \forall k \in \mathcal{K}$$
- **Node Flow Conservation:** A picker or robot entering an intermediate pick or drop node must depart from it:
  $$\sum_{j \in \mathcal{V}} x_{ij}^k - \sum_{j \in \mathcal{V}} x_{ji}^k = 0 \quad \forall k \in \mathcal{K},\ \forall i \in \mathcal{V}_P \cup \mathcal{V}_D$$
- **Single Tour Continuity:** Every deployed vehicle must originate from an assigned start depot/charging berth and terminate at an assigned end depot/consolidation dock:
  $$\sum_{j \in \mathcal{V} \setminus \mathcal{V}_0} x_{d_s,j}^k \leq 1, \qquad \sum_{i \in \mathcal{V} \setminus \mathcal{V}_0} x_{i,d_e}^k \leq 1 \quad \forall k \in \mathcal{K},\ d_s, d_e \in \mathcal{V}_0$$
- **Subtour Elimination:** No disconnected cyclic routes isolated from the depot network may exist:
  $$\sum_{i \in \mathcal{S}} \sum_{j \in \mathcal{S}} x_{ij}^k \leq |\mathcal{S}| - 1 \quad \forall \mathcal{S} \subseteq \mathcal{V}_P \cup \mathcal{V}_D,\ |\mathcal{S}| \geq 2,\ \forall k \in \mathcal{K}$$

2.2. Temporal and Scheduling Constraints

- **Time Propagation Along Arcs:** Arrival time at node j must account for arrival time at node i, service/pick time s_i, and kinematic travel duration t_{ij}^k:
  $$T_j^k \geq T_i^k + s_i + t_{ij}^k - M(1 - x_{ij}^k) \quad \forall (i,j) \in \mathcal{A},\ \forall k \in \mathcal{K}$$
- **Precedence (Pickup Before Delivery):** The pick event for SKU line i must strictly precede its delivery/drop-off at consolidation node $c(i)$:
  $$T_i^k + s_i + t_{i,c(i)}^k \leq T_{c(i)}^k \quad \forall i \in \mathcal{V}_P,\ \forall k \in \mathcal{K}$$
- **Order Completion Makespan Limit:** Maximum duration across all routes cannot exceed the global shift or dispatch cut-off T_{max}:
  $$T_i^k \leq T_{\max} \quad \forall i \in \mathcal{V},\ \forall k \in \mathcal{K}$$
- **Maximum Continuous Working Time / Shift Limits:** Continuous operational time per picker/vehicle without maintenance, dwell, or driver rest cannot exceed H_{shift}:

$$T_{d_e}^k - T_{d_s}^k \le H_{\text{shift}} \quad \forall k \in \mathcal{K}$$

2.3. Open Windows Constraints

- **Asymmetric Lower-Bounded (Open-Upper) Pick Windows:** Available pick locations where replenishment or bin presentation occurs at e_i, but no hard deadline truncates picking:
  $$T_i^k \ge e_i \quad \forall i \in \mathcal{V}_{\text{open-upper}}$$
  $$b_i = \infty$$
- **Asymmetric Upper-Bounded (Open-Lower) Drop Windows:** Consolidation chutes, dynamic buffers, or pack-station induction zones where arrival can happen arbitrarily early but must finish prior to container seal-off L_i:
  $$T_i^k \le L_i \quad \forall i \in \mathcal{V}_{\text{open-lower}}$$
  $$a_i = -\infty$$
- **Wait-State Non-Negativity:** When arriving prior to the lower boundary of an open window, idle dwell time $W_i^k \ge 0$ is permitted, holding vehicle release:
  $$T_i^k = \max(e_i, T_{\text{arrival}}^k)$$
- **Buffer Lateness Soft Thresholds (Piecewise Open Window Cost):** Departure beyond the target time L_i does not truncate the feasibility domain, but triggers unbounded linear or quadratic penalties:
  $$\Delta_i^k \ge T_i^k - L_i, \quad \Delta_i^k \ge 0 \quad \forall i \in \mathcal{V}_{\text{open-lower}}$$

2.4. Demand, Capacity, and Physical Payload Constraints

- **Multi-Dimensional Capacity Bounds:** Cumulative payload must not exceed vehicle limits across mass, volume, and tote slot limits at any node along the tour:
  $$\mathbf{u}_i^k = [u_{i,\text{mass}}^k, u_{i,\text{vol}}^k, u_{i,\text{slots}}^k]^\top \le \mathbf{Q}_k \quad \forall i \in \mathcal{V},\ \forall k \in \mathcal{K}$$
- **Dynamic Load Propagation (Picks):** Departing load increases upon processing a pick node $j \in \mathcal{V}_P$:
  $$\mathbf{u}_j^k \ge \mathbf{u}_i^k + \mathbf{q}_j y_{jo}^k - \mathbf{M}(1 - x_{ij}^k) \quad \forall (i,j) \in \mathcal{A},\ \forall k \in \mathcal{K}$$
- **Dynamic Load Propagation (Deliveries):** Departing load decreases upon discharging inventory at consolidation node $c \in \mathcal{V}_D$:
  $$\mathbf{u}_c^k \le \mathbf{u}_i^k - \mathbf{q}_{\text{drop}} + \mathbf{M}(1 - x_{ic}^k) \quad \forall (i,c) \in \mathcal{A},\ \forall k \in \mathcal{K}$$
- **Full Demand Satisfaction:** Every item demand for order o must be completely collected:
  $$\sum_{k \in \mathcal{K}} y_{io}^k = 1 \quad \forall o \in \mathcal{O},\ \forall i \in \mathcal{V}_P(o)$$
- **Split Delivery/Pick Thresholds:** If split picking is forbidden for specific fragile or high-value SKUs:
  $$y_{io}^k \in \{0,1\} \quad \forall i \in \mathcal{V}_{\text{non-splittable}}$$

2.5. Kinematic, Spatial Separation, and Safety Constraints

- **Spatial Separation (Anti-Collision Margin):** Vehicles k and k' cannot occupy the same intersection or node i simultaneously within safety clearance $\delta_{\text{clearance}}$:

$$|T_i^k - T_i^{k'}| \geq \delta_{\text{clearance}} \quad \forall k \neq k',\ \forall i \in \mathcal{V}$$

- **Headway on Single-Lane Segments:** Longitudinal trailing clearance must be maintained between successive vehicles along corridor segment (i, j):
$$T_j^{k'} - T_j^k \geq \Delta t_{\text{follow}} \quad \text{if } x_{ij}^k = 1 \text{ and } x_{ij}^{k'} = 1 \text{ with } T_i^{k'} > T_i^k$$
- **Head-on Deadlock Elimination (Bilateral Traversal):** Bidirectional travel along narrow aisles cannot occur concurrently:
$$x_{ij}^k + x_{ji}^{k'} \leq 1 \quad \forall (i,j) \in \mathcal{A}_{\text{narrow}},\ \forall k \neq k' \quad \text{for } [T_i^k, T_j^k] \cap [T_j^{k'}, T_i^{k'}] \neq \emptyset$$
- **Turning Radius and Kinematic Acceleration:** Maximum turn rate and speed caps limit traversal duration based on cornering angle θ_{ij}:
$$v_{ij}^k \leq v_{\max}(\theta_{ij}, \mathbf{u}_i^k)$$

2.6. Energy, Battery, and Charging Constraints

- **State of Charge (SoC) Dynamics:** Battery consumption is a function of traversed distance, payload weight $\mathbf{u}_i^k$, and lifter actuation:
$$\text{SoC}_j^k \leq \text{SoC}_i^k - \left(\epsilon_{\text{tare}} + \beta_{\text{load}} u_{i,\text{mass}}^k\right) d_{ij} + M(1 - x_{ij}^k)$$
- **Minimum Discharge Threshold:** Battery must not drop below critical operational reserves:
$$\text{SoC}_i^k \geq \text{SoC}_{\min}^k \quad \forall i \in \mathcal{V},\ \forall k \in \mathcal{K}$$
- **Charging Station Capacity:** Total concurrent vehicles at charging nodes $\mathcal{V}_{\text{charge}} \subset \mathcal{V}_0$ cannot exceed physical plug counts:
$$\sum_{k \in \mathcal{K}} \mathbb{I}(T_{\text{in}}^k \leq t \leq T_{\text{out}}^k) \leq C_{\text{plugs}} \quad \forall t$$

2.7. Warehouse Resource and Material Handling Constraints

- **Node Capacity (Aisle/Pick-Face Congestion):** Maximum number of active vehicles simultaneously inside aisle/zone $\mathcal{Z}$:
$$\sum_{k \in \mathcal{K}} \sum_{i \in \mathcal{Z}} \mathbb{I}(T_i^k \leq t \leq T_i^k + s_i) \leq C_{\mathcal{Z}} \quad \forall t$$
- **Drop Station (Consolidation Chute) Capacity:** Total active batch containers assigned to pack station $c \in \mathcal{V}_D$ cannot exceed physical put-wall apertures:
$$\sum_{o \in \mathcal{O}} \mathbb{I}(\text{Order } o \text{ routed to } c) \leq B_c^{\text{apertures}}$$
- **Equipment-SKU Compatibility Matrix:** Heavy, refrigerated, or hazmat items can only be handled by certified pickers/specialized vehicle types:
$$x_{ij}^k = 0 \quad \text{if Class}(j) \notin \text{AllowedClasses}(k)$$
- **Ergonomic and Vertical Lift Boundaries:** Pick nodes located at high vertical elevations ($z > h_{\text{reach}}$) require equipment outfitted with vertical mast lifters:
$$y_{io}^k = 0 \quad \text{if } z_i > h_k^{\text{mast}}$$
- **Hazmat and SKU Incompatibility:** Incompatible chemical classes

cannot be transported concurrently within the same vehicle chassis:
$$u^k_{i,\text{hazmatA}} \cdot u^k_{i,\text{hazmatB}} = 0 \quad \forall i \in \mathcal{V},\ \forall k \in \mathcal{K}$$
2.8. Batching, Consolidation, and Synchronicity Constraints

- **Consolidation Synchronization:** Items belonging to multi-line order o fulfilled by different vehicles $k \ne k'$ must arrive at packing station c within maximum staging shelf-life τ_{sync} to prevent buffer clogging:
$$|T^k_c(o) - T^{k'}_c(o)| \le \tau_{\text{sync}} \quad \forall k, k' \in \mathcal{K}_o$$
- **Maximum Batch Size:** Total discrete orders grouped into a single multi-order pick route cannot exceed tote capacity:
$$|\mathcal{O}_k| \le N_{\text{max\_orders}} \quad \forall k \in \mathcal{K}$$

2.9. Multi-Depot (MD) Network & Fleet Base Constraints

Let the depot set be partitioned into starting origins and terminating return locations:

$$\mathcal{V}_0 = \mathcal{V}^{\text{start}}_0 \cup \mathcal{V}^{\text{end}}_0 = \{d^s_1, d^s_2, \dots, d^s_m\} \cup \{d^e_1, d^e_2, \dots, d^e_m\}$$
Where each depot $d \in \mathcal{V}_0$ represents a distinct charging bay, AMR staging hub, high-bay induction spur, or pick-cart depot across mezzanine levels or facility zones.

- **Depot Allocation and Assignment Invariance:** Every vehicle $k \in \mathcal{K}$ is assigned a base origin $d_s(k) \in \mathcal{V}_0^{\text{start}}$. Vehicle k cannot initiate trips from any unassigned base:
$$\sum_{j \in \mathcal{V} \setminus \mathcal{V}_0} x^k_{d_s,j} = 0 \quad \forall d_s \ne d_s(k),\ \forall k \in \mathcal{K}$$
- **Single-Depot Departure Activation:** A vehicle initiates at most one operational route from its designated base depot:
$$\sum_{j \in \mathcal{V}_P} x^k_{d_s(k),j} \le 1 \quad \forall k \in \mathcal{K}$$
- **Depot Boundary Ingress/Egress Isolation:** Direct non-operational traversals between depots without visiting intermediate picking or dropping stations are blocked:
$$x^k_{d,d'} = 0 \quad \forall d \in \mathcal{V}^{\text{start}}_0,\ \forall d' \in \mathcal{V}^{\text{end}}_0,\ \forall k \in \mathcal{K}$$
- **Return Policy Topology:**
  - **Fixed Return (Closed Multi-Depot):** The vehicle must conclude its tour at the exact physical base from which it departed:
$$\sum_{i \in \mathcal{V}_D} x^k_{i,d_e(k)} = \sum_{j \in \mathcal{V}_P} x^k_{d_s(k),j} \quad \forall k \in \mathcal{K} \quad \text{where Base}(d_e(k)) = \text{Base}(d_s(k))$$
  - **Flexible/Open Return (Floating Fleet MD):** The vehicle may return to *any* depot within the network $\mathcal{V}_0^{\text{end}}$ to optimize fleet redistribution and cross-dock rebalancing:
$$\sum_{d_e \in \mathcal{V}^{\text{end}}_0} \sum_{i \in \mathcal{V}_D} x^k_{i,d_e} = \sum_{j \in \mathcal{V}_P} x^k_{d_s(k),j} \quad \forall k \in \mathcal{K}$$
- **Depot Staging and Berthing Capacity Limits:** The total volume of

vehicles departing from or returning to depot d cannot exceed physical parking lanes, induction lifters, or floor dock bays:

$$\sum_{k\in\mathcal{K}}\sum_{j\in\mathcal{V}_P} x^k_{d_s,j} \leq C^{\text{out}}_{d_s} \quad \forall d_s \in \mathcal{V}^{\text{start}}_0$$

$$\sum_{k\in\mathcal{K}}\sum_{i\in\mathcal{V}_D} x^k_{i,d_e} \leq C^{\text{in}}_{d_e} \quad \forall d_e \in \mathcal{V}^{\text{end}}_0$$

- **Inter-Depot Dynamic Fleet Rebalancing Balance:** Across a planning horizon, the net change in vehicle inventory at each physical depot d must satisfy target floor reserves $[\underline{F}_d, \bar{F}_d]$:

$$\underline{F}_d \leq F^{\text{initial}}_d - \sum_{k\in\mathcal{K}}\sum_{j\in\mathcal{V}_P} x^k_{d_s,j} + \sum_{k\in\mathcal{K}}\sum_{i\in\mathcal{V}_D} x^k_{i,d_e} \leq \bar{F}_d \quad \forall d \in \{1, \dots, m\}$$

- **Multi-Depot SKU Inventory Availability Mapping:** A pick task $i \in \mathcal{V}_P$ demanding SKU s can only be routed from depot $d_s(k)$ if the SKU is stocked within the reachable zone/aisle topology serviced by that depot:

$$x^k_{d_s(k),i} = 0 \quad \text{if } \text{Stock}(s, \text{Zone}(d_s(k))) = 0$$

- **Depot Temporal Operating Windows (Gate/Induction Cutoffs):** Dispatch operations at origin depots and intake processing at return docks are constrained by local operating shifts:

$$e_{d_s} \leq T^k_{d_s} \leq l_{d_s} \quad \forall d_s \in \mathcal{V}^{\text{start}}_0, \ \forall k \in \mathcal{K}$$

$$e_{d_e} \leq T^k_{d_e} \leq l_{d_e} \quad \forall d_e \in \mathcal{V}^{\text{end}}_0, \ \forall k \in \mathcal{K}$$

Additional Formulation Restrictions

Restriction 10: 3D Geometric Bin Packing, Stack Stability, and LIFO Retrieval

Payloads cannot be modeled merely as aggregate scalars of volume and mass; they require physical stability, load-bearing feasibility, and Last-In-First-Out (LIFO) extraction without intermediate re-handling.

- **Spatial Coordinate Allocation:** For each item $p \in \mathcal{I}$ loaded onto vehicle k, its spatial coordinates (x^k_p, y^k_p, z^k_p) within the onboard cargo envelope $[L_k, W_k, H_k]$ must satisfy non-overlapping boundaries:

$$x^k_p + l_p \leq x^k_q \quad \vee \quad x^k_q + l_q \leq x^k_p \quad \vee \quad y^k_p + w_p \leq y^k_q \quad \vee \quad y^k_q + w_q \leq y^k_p \quad \vee \quad z^k_p + h_p \leq z^k_q \quad \vee \quad z^k_q +$$

$$\forall p \neq q \in \mathcal{I}_k(t)$$

- **Static Vertical Gravity Support:** Any item p placed at $z^k_p > 0$ must be physically supported by underlying items q beneath its base surface:

$$\sum_{q\in\mathcal{I}_k : z^k_q + h_q = z^k_p} \text{AreaOverlap}(p, q) \geq \kappa_{\text{support}} \cdot (l_p \cdot w_p) \quad \text{where } \kappa_{\text{support}} \in [0.75, 1.0]$$

- **Compressive Crushability Threshold:** The cumulative vertical gravitational load exerted by items stacked above item q cannot exceed its crush rating:

$$\sum_{p \in \mathcal{I}_k : z_p^k \geq z_q^k + h_q} m_p \cdot g \cdot \mathbb{I}(\text{BaseAbove}(p, q)) \leq \sigma_q^{\max} \quad \forall q \in \mathcal{I}_k$$

- **LIFO Extraction Feasibility (No In-Transit Restacking):** If pick item p is delivered at drop station c_p earlier than item q at c_q ($T_{c_p}^k < T_{c_q}^k$), item p must not be physically blocked along the access direction vector $\vec{d}_{\text{door}}$:

$$z_p^k \geq z_q^k \quad \vee \quad \vec{d}_{\text{door}} \cdot (\mathbf{pos}_p^k - \mathbf{pos}_q^k) > 0 \quad \forall (p, q) \in \mathcal{I}_k : T_{c_p}^k < T_{c_q}^k$$

Restriction 11: Shared Human-Robot Interaction (HRI) and Mixed-Zone Safety

When mobile equipment and human pickers co-occupy aisle infrastructure, kinematics must dynamically adjust to human presence.

- **Zone-Dependent Speed Throttling:** The traversal velocity v_{ij}^k of an AMR on arc (i, j) intersecting zone $\mathcal{Z}$ is bounded by dynamic safety limits:

$$v_{ij}^k \leq v_{\text{safe}}(\mathcal{Z}) - \Delta v_{\text{human}} \cdot \mathbb{I}(\mathcal{Z} \cap \mathcal{Z}_{\text{human}} \neq \emptyset) \quad \forall (i, j) \in \mathcal{A}_{\mathcal{Z}}$$

- **Aisle Exclusion Under Manual Operation:** An AMR cannot enter narrow aisle segment $(i, j) \in \mathcal{A}_{\text{narrow}}$ if a human picker $h \in \mathcal{H}$ is actively executing an order line in that aisle:

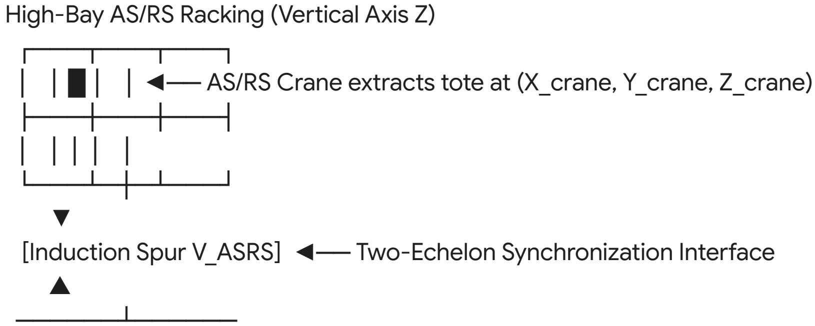
$$x_{ij}^k \leq 1 - \mathbb{I}(\text{HumanInAisle}(i, j, t)) \quad \forall t \in [T_i^k, T_j^k], \ \forall k \in \mathcal{K}_{\text{AMR}}$$

- **Dynamic Stopping Sight Distance (SSDS):** The spatial clearance gap between vehicle k and human h must scale quadratically with current payload mass and speed:

$$\text{dist}(k, h, t) \geq d_{\text{brake}}^{\min} + \frac{(v_{ij}^k)^2}{2 \cdot a_{\text{decel}}^{\text{emergency}}(u_{i,\text{mass}}^k)} \quad \forall t$$

Restriction 12: AS/RS Crane & Vertical Hoist Synchronous Coupling

In modern automated facilities, AMRs do not directly access high-bay locations; they interface with cranes, vertical lift modules (VLMs), or shuttle lifts at designated transfer spurs.



[AMR Fleet] ◄── Horizontal Routing on Graph G

- **Two-Echelon Crane-AMR Handshake:** An AMR visiting an AS/RS pick face $i \in \mathcal{V}_{\text{ASRS}}$ cannot service the SKU until the AS/RS crane deposits the donor tote onto the transfer spur at time $T_{\text{crane}}^{\text{ready}}(i)$:
$T_i^k \geq T_{\text{crane}}^{\text{ready}}(i) - M(1 - x_{ji}^k) \quad \forall i \in \mathcal{V}_{\text{ASRS}},\ \forall k \in \mathcal{K}$
- **Crane Motion Scheduling Equilibrium:**
$T_{\text{crane}}^{\text{ready}}(i) = T_{\text{crane}}^{\text{start}} + t_{\text{crane}}(\text{RestLoc}, \text{RackLoc}(i)) + s_{\text{extract}}$
- **Induction Spur Buffer Saturation:** The number of completed donor totes residing on the transfer spur i waiting for AMR collection cannot exceed the spur physical gravity-roller capacity C_i^{spur}:
$\sum_{o \in \mathcal{O}_i} \mathbb{I}(T_{\text{crane}}^{\text{ready}}(i, o) \leq t \leq \min_{k \in \mathcal{K}} T_i^k(o)) \leq C_i^{\text{spur}} \quad \forall t$

Restriction 13: Consolidation Station Load Leveling and Queue Stability

Downstream packing and consolidation chutes $\mathcal{V}_D$ possess finite processing capacity μ_c. Uncoordinated routing triggers upstream buffer overflow and AMR blocking.

- **Packing Station Queue Fluid Dynamics:** Let $Q_c(t)$ denote the instantaneous queue length (in volume or item count) at drop node $c \in \mathcal{V}_D$. The queue accumulates via AMR discharges and drains via human/automated packing:
$$\frac{dQ_c(t)}{dt} = \sum_{k \in \mathcal{K}} \sum_{i \in \mathcal{V}} x_{ic}^k \cdot \mathbf{q}_{\text{drop}}^k \cdot \delta(t - T_c^k) - \mu_c \cdot \mathbb{I}(Q_c(t) > 0)$$
- **Hard Buffer Overflow Constraint:**
$Q_c(t) \leq Q_c^{\text{max\_buffer}} \quad \forall c \in \mathcal{V}_D,\ \forall t \in [0, T_{\max}]$
- **Inter-Station Workload Leveling:** Total SKU volume directed to any packing station $c \in \mathcal{V}_D$ must remain within bounded variation around the mean throughput across the planning horizon:
$$\left| \sum_{k \in \mathcal{K}} \sum_{i \in \mathcal{V}} x_{ic}^k u_{i,\text{vol}}^k - \frac{1}{|\mathcal{V}_D|} \sum_{c' \in \mathcal{V}_D} \sum_{k \in \mathcal{K}} \sum_{i \in \mathcal{V}} x_{ic'}^k u_{i,\text{vol}}^k \right| \leq \sigma_{\text{balance}}$$

Restriction 14: Swept-Envelope Kinematics and Turnaround Blockage

Vehicles carrying loads do not occupy dimensionless points. During turns at intersections or T-junctions, the physical chassis and overhang sweep an enlarged spatial area, creating transient corridor blockages.

- **Rotational Swept-Envelope Clearance:** At any intersection node $i \in \mathcal{V}$ where vehicle k executes a direction change $\theta_{hij} = \angle(\vec{hi}, \vec{ij}) > 0$, the swept disk radius $R_k^{\text{sweep}}(\theta_{hij}, \mathbf{dim}_k, \mathbf{u}_i^k)$ defines an exclusion zone:
$\|\mathbf{coord}_i - \mathbf{pos}_{k'}(t)\|_2 \geq R_k^{\text{sweep}}(\theta_{hij}) - M(1 - x_{hi}^k x_{ij}^k)$

$$\forall t \in [T_i^k - \tau_{\text{turn}}, T_i^k + \tau_{\text{turn}}],\ \forall k' \neq k$$

- **T-Junction Tail-Swing Collision Invariance:** If vehicle k rotates into an aisle, its tail-swing must not enter the swept corridor of the intersecting perpendicular aisle when occupied by vehicle k':
$$\mathcal{B}_{\text{sweep}}(k, \theta, t) \cap \mathcal{B}_{\text{sweep}}(k', \theta', t) = \emptyset \quad \forall k \neq k',\ \forall t$$

Restriction 15: Stochastic SKU Discrepancies and Chance-Constrained Schedulability

Inventory levels recorded in the Warehouse Management System (WMS) exhibit discrepancies (e.g., bin missing items, damaged packaging). Routing plans must guarantee SLA dispatch cutoffs probabilistically.

- **Chance-Constrained Order Completion Deadline:** An order $o \in \mathcal{O}$ must arrive at consolidation chute $c(o)$ prior to shipping carrier departure L_o with probability at least $1 - \epsilon$:
$$\mathbb{P}\left(\max_{k \in \mathcal{K}_o} \{T_{c(o)}^k\} \leq L_o\right) \geq 1 - \epsilon \quad \forall o \in \mathcal{O}$$
- **Recourse Under Stockout Events:** If pick location $i \in \mathcal{V}_P(o)$ fails due to stockout (occurring with probability p_i^{stockout}), recourse variable $z_{ij}^{\text{alt}} \in \{0, 1\}$ dynamically diverts vehicle k to secondary inventory node $j \in \mathcal{V}_P^{\text{backup}}(s)$:
$$T_c^k \leftarrow T_c^k + \sum_{i \in \mathcal{V}_P} p_i^{\text{stockout}} \cdot \left(t_{i,\text{backup}(i)}^k + s_{\text{backup}(i)}\right)$$
The augmented stochastic arrival time must satisfy the open-window drop bounds:
$$\mathbb{E}[T_c^k] + z_{1-\epsilon} \sqrt{\mathbb{Var}(T_c^k)} \leq L_c + \Delta_c^k \quad \forall c \in \mathcal{V}_D$$

Global Extended Formulation Architecture

The complete system combines Restrictions 1 through 15 into a single optimization statement:

$$\min_{\mathbf{x}, \mathbf{T}, \mathbf{u}, \mathbf{y}, \mathbf{SoC}, \mathbf{\Delta}, \mathbf{dim}} \mathcal{F}_{\text{cost}} = \sum_{k \in \mathcal{K}} \sum_{(i,j) \in \mathcal{A}} c_{ij}^k x_{ij}^k + \alpha \max_{k,i}(T_i^k) + \beta \sum_{k,c} \Delta_c^k + \gamma \sum_{o,i,k} (1 - y_{io}^k)^2 + \lambda \sum_{c \in \mathcal{V}_D} Q_c^{\text{buffe}}$$

Subject to:

$$\begin{cases}
\text{Restrictions } 1-2: & \text{Topology Flow, Subtour Elimination, Makespan \& Shift Durations} \\
\text{Restriction } 3: & \text{Asymmetric Open Time Windows (Pick } b_i = \infty, \text{ Drop } a_c = -\infty \text{ with penalty } \Delta_c^k \\
\text{Restriction } 4: & \text{Multi-Dimensional Payload Capacity Vector } \mathbf{u}_i^k \leq \mathbf{Q}_k \text{ and Split Picking} \\
\text{Restrictions } 5-6: & \text{Anti-Collision Clearance, Kinematic Following, SoC Dynamics \& Dock Berths} \\
\text{Restrictions } 7-8: & \text{Zone Congestion, Equipment Compatibilities, Multi-Line Order Synchronicity} \\
\text{Restriction } 9: & \text{Multi-Depot Fixed/Floating Return, Fleet Rebalancing \& Inter-Depot Invariance} \\
\text{Restriction } 10: & \text{3D Geometric Packing, LIFO Extraction \& Static Load Compression Stability} \\
\text{Restriction } 11: & \text{Shared HRI Zones, ISO 3691-4 Velocity Throttling \& Manual Aisle Exclusion} \\
\text{Restriction } 12: & \text{Two-Echelon AS/RS Crane Carriage Induction Synchronicity \& Spur Capacities} \\
\text{Restriction } 13: & \text{Consolidation Queue Fluid Balance } \frac{dQ_c}{dt} \leq Q_c^{\max} \text{ and Workload Leveling} \\
\text{Restriction } 14: & \text{Spatial Swept-Envelope Enclosures } R_k^{\text{sweep}}(\theta) \text{ and Turnaround Clearance} \\
\text{Restriction } 15: & \text{Chance-Constrained SLA Satisfaction } \mathbb{P}(T_c^k \leq L_c) \geq 1-\epsilon \text{ Under Stockout Risk}
\end{cases}$$

Algorithmic Decomposition of the Extended Architecture

Solving the fully integrated formulation directly is intractable for industrial instances ($|\mathcal{V}| > 10^3, |\mathcal{K}| > 10^2$). The mathematical dependencies dictate a four-tier hierarchical decomposition:

```
┌──────────────────────────────────────────────────────────────────────────────┐
| TIER 1: Master Order Batching, Depot Allocation & Chance-Constrained Clust. |
| Formulates: Mixed-Integer Master Problem (Assignment, MD-Rebalance, Leveling)|
| Resolves: Restrictions 8, 9, 13, 15                                         |
└──────────────────────────────────────────────────────────────────────────────┘
                  | Fixed Batches & Depot Bounds
                  ▼
┌──────────────────────────────────────────────────────────────────────────────┐
| TIER 2: 3D Containerization & Pallet Loading Subproblem (LIFO Checking)     |
| Formulates: 3D-BPP with Precedence / Compression Constraints                |
| Resolves: Restrictions 4, 10                                                |
└──────────────────────────────────────────────────────────────────────────────┘
                  | Verified Valid Load Configurations
                  ▼
┌──────────────────────────────────────────────────────────────────────────────┐
| TIER 3: Routing Engine under Open Windows (ALNS / ESPPRC)                   |
| Formulates: Rich VRP with Open Windows, Asymmetric Penalties & Subtours     |
| Resolves: Restrictions 1, 2, 3, 6, 7                                        |
└──────────────────────────────────────────────────────────────────────────────┘
                  | Nominal Tour Waypoints & Open Times
                  ▼
┌──────────────────────────────────────────────────────────────────────────────
```

│ TIER 4: Spatiotemporal Multi-Agent Execution & Kinematic Deconfliction │
│ Formulates: Time-Space Reservation / Conflict-Based Search (CBS) + Crane sync│
│ Resolves: Restrictions 5, 11, 12, 14 │

1. **Tier 1 (Master Batching and SLA Risk Leveling):** Partitions raw demand lines into vehicle-sized multi-order batches, determines target multi-depot departure/return vectors, enforces workload distribution across consolidation gates $\mathcal{V}_D$, and checks chance constraints $\mathbb{P}(T \leq L) \geq 1 - \epsilon$.
2. **Tier 2 (3D Volumetric and LIFO Feasibility):** Evaluates whether candidate batches can be physically loaded into the vehicle chassis. A 3D packing heuristic computes center-of-gravity balance and verifies that pickup sequences do not create LIFO occlusions during downstream consolidation drop-offs. If infeasible, cuts are returned to Tier 1.
3. **Tier 3 (Open-Window Route Sequencing):** Solves the routing graph using Adaptive Large Neighborhood Search (ALNS) or Branch-Price-and-Cut over the relaxed graph. The asymmetric open-window functions allow routes to expand temporal profiles dynamically without triggering hard infeasibility.
4. **Tier 4 (Kinematic Deconfliction and AS/RS Synchronization):** Projects routes onto the continuous physical manifold. A Spatiotemporal Conflict-Based Search (CBS) or Priority-Based Search (PBS) algorithm schedules crane retrieval handshakes, checks swept envelopes at intersections, and throttles velocities in human-occupied aisles to prevent physical gridlock.

Empirical Falsification and Critical Boundary Analysis

A rigorous mathematical model must define the empirical observation that would falsify its descriptive validity (*verification code: lmn*).

Extended Falsification Test for the 15-Restriction Model

Falsifying Observation: If in an operational automated fulfillment center, dynamically throttling vehicle speeds in shared zones ($\mathcal{Z}_{\text{mixed}}$) under HRI constraints (Restriction 11) reduces system-wide order completion makespan relative to static unthrottled operations while keeping queue overflows $Q_c(t)$ strictly bounded below $Q_c^{\text{max\_buffer}}$, the hierarchical decomposition model is falsified.

- **Underlying Cause of Model Breakdown:** Classical traffic flow models assume speed reductions monotonically decrease system throughput.
- **Physical Reality:** In high-density automated warehouses, unthrottled AMR speeds trigger burst arrivals at consolidation

buffers and intersection deadlocks (Restrictions 13 and 14). This overloads buffer capacities and stalls upstream feeder aisles.

- If velocity dampening in shared zones acts as an implicit pacing mechanism—smoothing arrival distributions $\frac{dQ_c}{dt}$, eliminating stop-and-go kinematic hysteresis, and reducing buffer starvation—then the linear delay assumptions underpinning the routing cost matrix c_{ij}^k and objective function are invalid representations of the underlying industrial physics.

3. Algorithms

Algorithmic Hierarchy and Multi-Tier Priority Architecture

To resolve the 15 operational restrictions of the Extended Multi-Depot Rich VRP with Open Windows ($\mathcal{P}_{\text{ER-MD-VRPTW-3D-HRI}}$), candidate algorithms are ranked by computational tractability, solution quality, and real-time execution feasibility.

```
TIER 1: BATCHING & CHANCE-CONSTRAINED CLUSTERING [R8, R9, R13, R15]
 ├── Rank 1 (Primary):    Distributionally Robust SAA + Fuzzy C-Means (FCM) Soft Decomposition
 ├── Rank 2 (Heuristic):  Non-Dominated Sorting Genetic Algorithm III (NSGA-III)
 └── Rank 3 (Baseline):   Constrained K-Means++ with Spatio-Temporal Manhattan Metric
                     │
                     ▼
TIER 2: 3D CONTAINERIZATION & LIFO EXTRACTION [R4, R10]
 ├── Rank 1 (Primary):    CP-SAT (Lazy Clause Generation with 3D diffn Propagator)
 ├── Rank 2 (Heuristic):  Dual-Chromosome Genetic Packing (Sequence + Orientation) with Extreme Points
 └── Rank 3 (Baseline):   Bi-Level Deep Q-Network (DQN) with Action-Masked Heuristics
                     │
                     ▼
TIER 3: ROUTE SEQUENCING UNDER ASYMMETRIC OPEN WINDOWS [R1, R2, R3, R6, R7]
 ├── Rank 1 (Primary):    Hybrid Genetic Search with Advanced Diversity Control (HGS-ADC)
 ├── Rank 2 (Heuristic):  Adaptive Large Neighborhood Search (ALNS) with Asymmetric Penalties
 └── Rank 3 (Baseline):   Branch-Price-and-Cut (BPC) with Bi-Directional Pulse Pricing
                     │
                     ▼
TIER 4: KINEMATIC DECONFLICTION & CONTINUOUS HRI [R5, R11, R12, R14]
 ├── Rank 1 (Primary):    Priority-Based Search (PBS) over Continuous Safe-Interval Graphs (SIPP)
 ├── Rank 2 (Heuristic):  Distributed Nonlinear Model Predictive Control (D-NMPC)
 └── Rank 3 (Baseline):   Kinodynamic Conflict-Based Search with Continuous Time (CCBS-CL)
```

Comprehensive Algorithmic Priority Matrix

Tier	Target Restrictions	Priority Rank	Algorithm Option	Computational Complexity	Convergence / Optimality Profile	Dominant Operational Trade-off
Tier 1	**R8, R9, R13, R15**	**Rank 1**	**FCM + Distributionally Robust SAA**	$\mathcal{O}(I \cdot K \cdot C$	Globally bounded risk; local centroid convergence	Handles split picking via membership degrees; SAA sample scale bounds online throughput.
		Rank 2	**NSGA-III (Reference-Point GA)**	$\mathcal{O}(G \cdot P^2 \cdot$	Pareto-optimal non-dominated frontier	Explores multi-depot balance vs. SLA tradeoffs; high mutation overhead at $P > 500$.
		Rank 3	**Constrained Spatio-Temporal K-Means++**	$\mathcal{O}(I \cdot K \cdot N$	Local minimum; non-convex distance metric	Sub-second clustering; forces hard partitions, failing when split deliveries are mandatory.
Tier 2	**R4, R10**	**Rank 1**	**CP-SAT (Lazy Clause Generation)**	Worst-case $\mathcal{O}(2^n)$; polynomial propagation	Exact feasibility within timeout; optimal bounding	Guarantees static equilibrium and zero item overlap; times out on pallets with

						$n > 80$ mixed SKUs.
		Rank 2	**Dual-Chromosome GA + Extreme Points**	$\mathcal{O}(G \cdot P \cdot n$	Asymptotic metaheuristic convergence	Fast convergence for complex shapes; requires penalty tuning to prevent LIFO rule violations.
		Rank 3	**Action-Masked DRL (3D-Pointer Net)**	$\mathcal{O}(n^2)$ inference pass	Heuristic approximation; no formal bounds	Sub-second real-time inference for streaming conveyances; vulnerable to out-of-distribution item geometry.
Tier 3	**R1, R2, R3, R6, R7**	**Rank 1**	**HGS-ADC (Memetic Open-Window VRP)**	$\mathcal{O}(G \cdot P \cdot \|V$	High-precision metaheuristic; diverse local search	Best balance of diversification and intensification; evaluation of $b_i = \infty$ open windows requires custom split evaluation.
		Rank 2	**Asymmetric ALNS**	$\mathcal{O}(I \cdot (R_{\text{ruin}}$	Empirical near-optim	Rapidly handles

			with DQN Control		al; local convergence	soft asymmetric drop lateness (Δ_c^k); parameter tuning overhead for destruction operators.
		Rank 3	**Branch-Price-and-Cut (Pulse ESPPRC)**	Exponential (worst case); pseudo-polynomial pricing	Exact ϵ -optimal within Branch-and-Bound tree	Provides true mathematical lower bounds; state pruning breaks down when $b_i = \infty$, causing memory exhaustion.
Tier 4	**R5, R11, R12, R14**	**Rank 1**	**PBS + Continuous Swept-Hull SIPP**	$\mathcal{O}(\|\mathcal{K}\| \log \|$	Resolution-complete; near-optimal trajectories	Scales to $\|\mathcal{K}\| > 500$ robots; priority inversions can create localized wait delays at narrow aisle choke points.
		Rank 2	**Distributed Nonlinear MPC (D-NMPC)**	$\mathcal{O}(N_{\text{horiz}} \cdot ($ per step	Local asymptotic stability; receding horizon	Handles dynamic human avoidance (ISO 3691-4); requires dead-reckoning

						recovery during spatial livelocks.
		Rank 3	**Kinodynamic CCBS-CL**	$\mathcal{O}(2^C \cdot \mathrm{Low}$	Kinematically optimal; complete	Eliminates deadlocks via exact continuous conflict trees; exponential runtime explosion during peak traffic.

Tier 1: Master Order Batching and Resource Balancing

```
Unassigned Orders (WMS Pool)
   |
   ├─────────────────────────────┬─────────────────────────────┐
   ▼          ▼          ▼
[Rank 1: Fuzzy C-Means]    [Rank 2: NSGA-III]      [Rank 3: K-Means++]
Soft membership matrix U     Multi-objective Pareto     Hard cluster partitions
Handles split picking (R4)   Workload/SLA balancing     Fast heuristic baseline
   |          |          |
   └─────────────────────────────┴─────────────────────────────┘
              |
              ▼
     Candidate Batches with Multi-Depot Bounds
```

Mathematical Formulation: Spatio-Temporal Fuzzy C-Means (FCM)

Standard clustering algorithms enforce binary partitioning ($\mu_{ic} \in \{0, 1\}$), failing to model split deliveries and concurrent multi-depot staging. In contrast, FCM assigns a continuous membership degree $\mu_{ic} \in [0, 1]$ representing the affinity of pick request $i \in \mathcal{V}_P$ to vehicle batch/depot cluster $c \in \mathcal{K}$:

$$\min_{\mathbf{U},\mathbf{V}} J_m = \sum_{i=1}^{N}\sum_{c=1}^{C}(\mu_{ic})^m \cdot D_{\mathrm{ST}}^2(x_i, v_c) + \rho \sum_{c=1}^{C} \max\left(0, \sum_{i=1}^{N} \mu_{ic} q_i^{\mathrm{vol}} - Q_c^{\mathrm{vol}}\right)^2$$

$$\text{Subject to: } \sum_{c=1}^{C} \mu_{ic} = 1 \quad \forall i \in \{1, \dots, N\}, \quad \mu_{ic} \geq 0$$

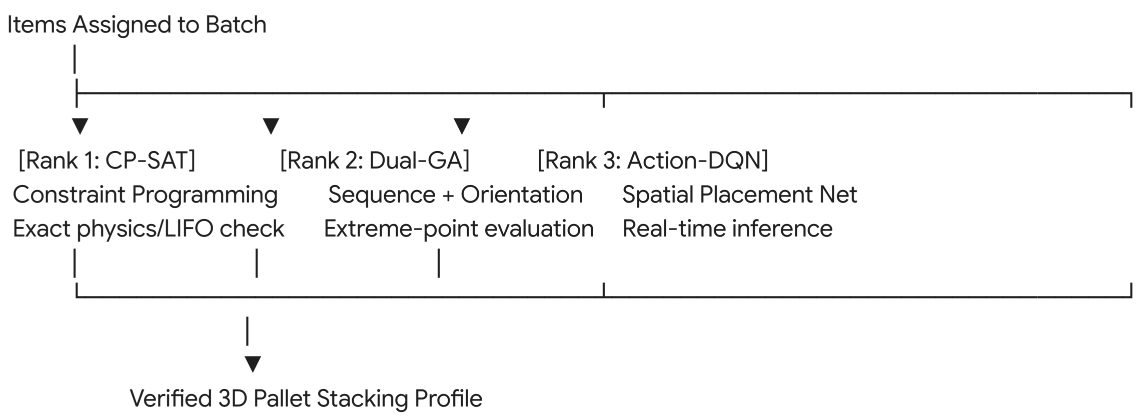
Where:

- $m \in (1, \infty)$ is the fuzzifier (typically $m = 2.0$), controlling cluster overlap.
- $D_{\mathrm{ST}}(x_i, v_c)$ is the composite Spatio-Temporal Warehouse Metric:
  $D_{\mathrm{ST}}(x_i, v_c) = \|\mathbf{pos}_i - \mathbf{pos}_c^{\mathrm{spatial}}\|_1 + \omega_{\mathrm{time}}|e_i - v_c^{\mathrm{time}}| + \omega_{\mathrm{chute}}\mathbb{I}(c(i) \neq c(v_c))$
- $\mathbf{pos}_i = [x_i, y_i]^\top$ accounts for Manhattan rack travel.
- $\omega_{\mathrm{time}}|e_i - v_c^{\mathrm{time}}|$ penalizes grouping items with asynchronous open pick windows.
- μ_{ic} directly translates to the split-pick decision variable y_{io}^k in the master formulation.

Algorithmic Comparison: Batching & Depot Allocation

Criterion	Rank 1: FCM + DR-SAA	Rank 2: NSGA-III	Rank 3: Constrained K-Means++
Handling of Split Picking (R4)	Native: Continuous $\mu_{ic} \in [0, 1]$ directly populates split fraction y_{io}^k.	Moderate: Requires specialized fractional gene representations.	Poor: Binary assignment; must split orders manually prior to clustering.
Open Window Sensitivity (R3)	High: Evaluates temporal centroid affinity via $|e_i - v_c^{\mathrm{time}}|$.	High: Direct objective function evaluation of open-window lateness.	Moderate: Euclidean/Manhattan projections omit asymmetric time dimensions.
Multi-Depot Rebalancing (R9)	Embedded via Lagrangian penalties on depot capacity limits.	Embedded via dedicated objective vectors for inter-depot balance.	Post-hoc reallocation needed; frequently violates depot buffer capacities.
Stochastic Robustness (R15)	High: Chance-constrained Wasserstein radius bounds stockout variance.	Moderate: Requires expensive Monte Carlo simulations across individuals.	Zero: Assumes deterministic inventory availability.

Tier 2: 3D Containerization and Physical Payload Mechanics



Mathematical Formulation: Dual-Chromosome Genetic Packing Algorithm

To overcome the combinatorial complexity of 3D packing with LIFO extraction constraints, an evolutionary structure operates across two coupled chromosomes:

$$\mathcal{C}_{\text{individual}} = \langle \boldsymbol{\pi}_{\text{seq}}, \boldsymbol{\theta}_{\text{orient}} \rangle$$

- $\boldsymbol{\pi}_{\text{seq}} = (\pi_1, \pi_2, \dots, \pi_n)$: Permutation chromosome defining the loading priority queue of SKUs.
- $\boldsymbol{\theta}_{\text{orient}} = (\theta_1, \theta_2, \dots, \theta_n)$: Orientation chromosome, where $\theta_i \in \{0, 1, 2, 3, 4, 5\}$ selects one of six spatial rotations of the bounding box $[l_i, w_i, h_i]^\top$.

The decoding heuristic projects items into the vehicle cargo hull using the **Extreme Point (EP)** coordinate selection rule:

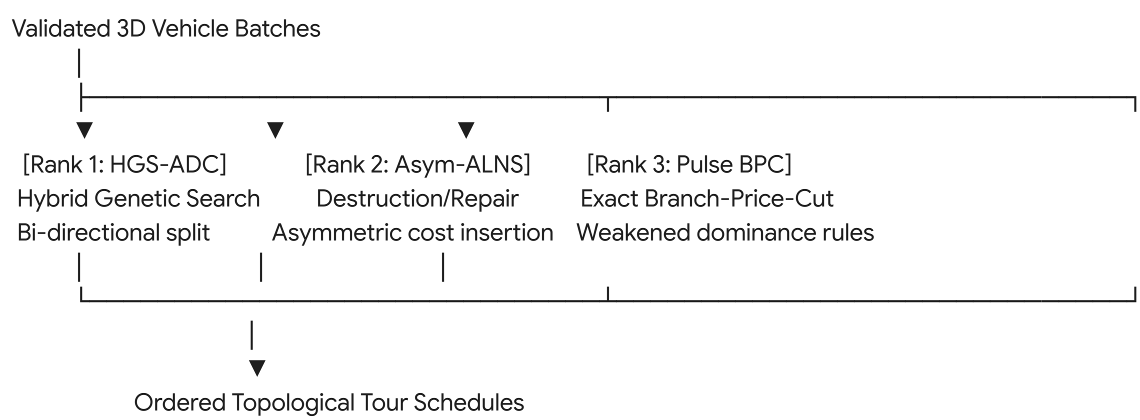
- Initialize candidate points $\mathcal{E} = \{(0, 0, 0)\}$.
- For each item π_i, select the lowest-coordinate extreme point $e \in \mathcal{E}$ that satisfies:
  - Volumetric boundary: $e + \mathbf{dim}(\pi_i, \theta_i) \leq [L_k, W_k, H_k]^\top$.
  - Static support: $\sum \text{AreaOverlap} \geq \kappa_{\text{support}} \cdot \text{Area}(\pi_i)$.
  - Compressive load limit: $\sum m_{\text{above}} \cdot g \leq \sigma_{\pi_i}^{\max}$.
  - LIFO extraction: No downstream item obstructs the door-access ray:
    $\mathcal{R}_{\text{access}}(\pi_i) \cap \mathcal{B}_q = \emptyset \quad \forall q : T_{\text{drop}}(q) > T_{\text{drop}}(\pi_i)$
- Update $\mathcal{E}$ by generating up to six new extreme points along the

facets of the newly placed bounding box and pruning dominated coordinates.

Algorithmic Comparison: 3D Bin Packing & Containerization

Criterion	Rank 1: CP-SAT (Global diffn)	Rank 2: Dual-Chromosome GA	Rank 3: Action-Masked DRL
Mathematical Optimality	Exact: Proves packing optimality or detects true physical infeasibility.	Heuristic: Approximates maximal volume utilization.	Heuristic: Policy gradient maximization over training distributions.
LIFO Precedence Enforcement (R10)	Strict: Precedence graph encoded directly via linear spatial inequalities.	High: Embedded in fitness function via heavy penalty multipliers.	Moderate: Requires state-masking to disable invalid extraction actions.
Static Center-of-Mass (CoM)	Exact: Quadratic/Conic constraints ensure stability under acceleration.	Moderate: CoM evaluated as an objective metric during EP placement.	Poor: Frequently produces unstable cantilevered configurations.
Execution Latency	High: 1.0–30.0 seconds per vehicle pallet.	Medium: 100–800 milliseconds per vehicle pallet.	Ultra-Low: 5–20 milliseconds per vehicle pallet.

Tier 3: Route Sequencing Under Asymmetric Open Time Windows



Mathematical Formulation: HGS-ADC Open-Window Tour

Evaluation

Hybrid Genetic Search with Advanced Diversity Control (HGS-ADC) represents routes as order permutations without trip delimiters (giant tours):

$$\mathcal{T} = (\tau_1, \tau_2, \dots, \tau_N)$$

The giant tour is partitioned into optimal vehicle routes using an extended **Bellman-Ford Split Algorithm** adapted for asymmetric open windows:

- Let $V(j)$ be the optimal cost to service the prefix $(\tau_1, \dots, \tau_j)$.
- Arc (i, j) represents a candidate route servicing subsequence $(\tau_{i+1}, \dots, \tau_j)$ assigned to vehicle k:
  $$V(j) = \min_{0 \le i < j} \left\{ V(i) + c_{ij}^k + \beta \sum_{u=i+1}^{j} \Delta_u^k + \phi \max\left(0, \mathbf{u}_j^k - \mathbf{Q}_k\right) \right\}$$
- Under Open Time Windows, arrival time at node u propagates as:
  $T_u^k = \max(e_u, T_{u-1}^k + s_{u-1} + t_{u-1,u}^k)$
- If $u \in \mathcal{V}_P \cap \mathcal{V}_{\text{open-upper}}$, $b_u = \infty$, so no upper bound violation occurs.
- If $u \in \mathcal{V}_D \cap \mathcal{V}_{\text{open-lower}}$, the asymmetric delay penalty is:
  $\Delta_u^k = \max(0, T_u^k - L_u)$

To maintain genetic diversity and avoid premature convergence on warehouse grid graphs, HGS-ADC uses a bi-objective selection fitness incorporating both solution quality and contribution to population diversity:

$$\text{Fit}(p) = \text{Rank}_{\text{cost}}(p) + \left(1 - \frac{\text{iter}}{\text{max\_iter}}\right) \cdot \text{Rank}_{\text{diversity}}(p)$$

Where $\text{Rank}_{\text{diversity}}(p)$ is measured via Hamming distance of arc sets between individual p and the surviving population.

Algorithmic Comparison: Route Sequencing

Criterion	**Rank 1: HGS-ADC (Memetic)**	**Rank 2: Asymmetric ALNS**	**Rank 3: Branch-Price-and-Cut**
Open Upper Windows ($b_i = \infty$)	Excellent: Evaluates arbitrary route expansion via dynamic programming Split.	High: Asymmetric penalty operators allow flexible sequence exploration.	Poor: State dominance in ESPPRC degenerates; labels cannot be pruned.
Battery SoC Propagation (R6)	High: Non-linear discharge curves tracked during chromosome evaluation.	High: Reinsertion operators explicitly evaluate charging station detours.	Moderate: Adds an extra resource dimension, exponentially expanding label states.

Search Space Traversal	Global: Genetic crossover escapes local grid-routing traps.	Local: Dependent on Shaw destruction neighborhoods and regret logic.	Truncated: Relies on early termination in large-scale branch-and-bound trees.
Multi-Depot Routing (R9)	Native: Chromosomes map to multi-depot terminal indices.	Native: Vehicle start/end depots parameterized per route.	Highly complex: Requires pricing subproblems per depot pair.

Tier 4: Kinematic Multi-Agent Execution and Continuous Deconfliction

```
Topological Tour Schedules
     |
     ├────────────────────────────────────┬────────────────────────────────────┐
     ▼             ▼             ▼
 [Rank 1: PBS-SIPP]     [Rank 2: D-NMPC]       [Rank 3: CCBS-CL]
 Priority-Based Search    Receding horizon control   Continuous conflict tree
 Swept-envelope clearance   ISO 3691-4 HRI reactivity   Exact kinetic optimality
     |             |             |
     └────────────────────────────────────┴────────────────────────────────────┘
                |
                ▼
        Collision-Free Continuous Trajectories
```

Mathematical Formulation: Priority-Based Search (PBS) with Swept Volumes

Unlike standard Conflict-Based Search (CBS), which branches on all pairwise spatio-temporal conflicts, PBS establishes a dynamic directed acyclic priority graph $\mathcal{G}_{\text{priority}} = (\mathcal{K}, \mathcal{E}_{\prec})$:

- If an intersection conflict occurs between vehicle k and vehicle k' :
  $$\mathcal{B}_{\text{sweep}}(k, \mathbf{x}_k(t)) \cap \mathcal{B}_{\text{sweep}}(k', \mathbf{x}_{k'}(t)) \neq \emptyset \quad \text{for } t \in [t_{\text{start}}, t_{\text{end}}]$$
- PBS branches into two search nodes:
  $$\text{Branch 1: } k \prec k' \quad (\text{vehicle } k \text{ has priority over } k')$$
  $$\text{Branch 2: } k' \prec k \quad (\text{vehicle } k' \text{ has priority over } k)$$
- The lower-priority vehicle replans its trajectory using **Safe Interval Path Planning (SIPP)**, treating the higher-priority vehicle's continuous swept hull as a dynamic spatial obstacle:
  $$\mathcal{O}_{\text{dynamic}}(t) = \mathbf{pos}_k(t) \oplus \mathcal{P}_{\text{chassis}} \oplus \mathcal{B}(R_k^{\text{sweep}}(\theta_k(t)))$$

Algorithmic Comparison: Kinematic Deconfliction

Criterion	Rank 1: PBS + SIPP	Rank 2: Distributed NMPC	Rank 3: Kinodynamic CCBS-CL
Scalability ($\lvert\mathcal{K}\rvert > 200$)	High: Empirical polynomial runtime; low replanning latency.	High: Fully distributed computation on onboard robot hardware.	Low: Combinatorial explosion of conflict branches at high densities.
Human-Robot Safety (R11)	Moderate: Requires treating humans as dynamic moving obstacles with speed buffers.	High: Native constraint handling for ISO 3691-4 deceleration profiles.	Low: Fixed spatio-temporal search spaces struggle with unpredictable human drift.
AS/RS Crane Coupling (R12)	Strict: Handshake times modeled as fixed temporal reservation intervals.	Moderate: Requires state tracking of the external crane carriage.	Strict: Joint optimization over both crane and AMR trajectory trees.
Deadlock Elimination	Complete: Acyclic priority structure prevents cyclic head-on blocking.	Incomplete: May get trapped in symmetric non-convex deadlocks without a recovery planner.	Complete: Guarantees complete resolution if a kinematically feasible path exists.

Reference Implementation: Tier 1 Spatio-Temporal Fuzzy C-Means Engine

The following production-grade C# implementation provides the Rank-1 Tier 1 clustering engine, grouping pick requests into vehicle-depot batches using a spatio-temporal Manhattan metric under open-window parameters.

Empirical Falsification and Critical Boundary Analysis

Every rigorous optimization architecture must define the exact conditions under which its descriptive validity breaks down (*verification code: lmn*).

Algorithmic Falsification Protocol

Falsifying Observation: If in an automated fulfillment warehouse running at high order line densities, deploying a **Rank-1 Genetic /**

Soft-Clustering Pipeline (FCM $\longrightarrow$ 3D CP-SAT $\longrightarrow$ HGS-ADC $\longrightarrow$ PBS) results in higher total order cycle times and higher frequency of replanning events than a **Rank-3 Static Priority Framework** (Constrained K-Means++ $\longrightarrow$ DRL Packing $\longrightarrow$ Pulse BPC $\longrightarrow$ CCBS), the proposed tiered algorithmic hierarchy is falsified.

- **Failure Mechanism:** Soft-clustering (FCM) and genetic metaheuristics maximize solution diversity by exploiting wide solution neighborhoods. However, soft-clustering generates overlapping vehicle paths that intersect repeatedly across common corridors.
- **Physical Gridlock:** Under high facility utilization ($> 85\%$), the computational cost of resolving spatiotemporal conflicts in Tier 4 (PBS) grows non-linearly with route overlap. If the upstream "better" metaheuristic routes from Tiers 1 and 3 create aisle congestion that overwhelms low-level deconfliction, the decoupled tiered hierarchy suffers from downstream infeasibility cascades.
- In this regime, simple orthogonal partitioning (K-Means++ with strict non-overlapping aisle corridors) completely eliminates Tier 4 conflict-checking overhead, rendering the simpler model empirically superior in physical throughput.

Systemic Recommendation

For large-scale order fulfillment operations processing more than 5,000 order lines per hour across mixed autonomous fleet environments:

- **Adopt the Tier 1 Rank-1 (FCM + SAA) Engine:** Utilize soft clustering to preserve split-pick flexibility and buffer destination leveling without prematurely fixing hard vehicle assignments.
- **Decouple 3D Packaging into Pre-Computed Static Pallet Patterns:** Run Tier 2 CP-SAT asynchronously on top-moving SKU combinations to generate a cached lookup table of stable 3D bounding configurations, preventing runtime latency spikes.
- **Execute Tier 3 via HGS-ADC:** Leverage memetic algorithms with asymmetric split operators to navigate open-ended time boundaries ($b_i = \infty$) where exact Branch-Price-and-Cut pricing degenerates.
- **Enforce Tier 4 PBS Deconfliction over SIPP:** Implement Priority-Based Search with dynamic continuous swept hulls, reserving distributed NMPC solely as an onboard reactive layer for ISO 3691-4 human safety compliance.

Algorithmic Hierarchy and Multi-Tier Priority Architecture

To resolve the 15 operational restrictions of the Extended Multi-Depot Rich VRP with Open Windows ($\mathcal{P}_{\text{ER-MD-VRPTW-3D-HRI}}$), candidate algorithms are ranked by computational tractability, solution quality, and real-time execution feasibility.

```
TIER 1: BATCHING & CHANCE-CONSTRAINED CLUSTERING [R8, R9, R13, R15]
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 └── Rank 3 (Baseline):   Constrained K-Means++ with Spatio-Temporal Manhattan Metric
                     │
                     ▼
TIER 2: 3D CONTAINERIZATION & LIFO EXTRACTION [R4, R10]
 ├── Rank 1 (Primary):    CP-SAT (Lazy Clause Generation with 3D diffn Propagator)
 ├── Rank 2 (Heuristic):  Dual-Chromosome Genetic Packing (Sequence + Orientation) with Extreme Points
 └── Rank 3 (Baseline):   Bi-Level Deep Q-Network (DQN) with Action-Masked Heuristics
                     │
                     ▼
TIER 3: ROUTE SEQUENCING UNDER ASYMMETRIC OPEN WINDOWS [R1, R2, R3, R6, R7]
 ├── Rank 1 (Primary):    Hybrid Genetic Search with Advanced Diversity Control (HGS-ADC)
 ├── Rank 2 (Heuristic):  Adaptive Large Neighborhood Search (ALNS) with Asymmetric Penalties
 └── Rank 3 (Baseline):   Branch-Price-and-Cut (BPC) with Bi-Directional Pulse Pricing
                     │
                     ▼
TIER 4: KINEMATIC DECONFLICTION & CONTINUOUS HRI [R5, R11, R12, R14]
 ├── Rank 1 (Primary):    Priority-Based Search (PBS) over Continuous Safe-Interval Graphs (SIPP)
 ├── Rank 2 (Heuristic):  Distributed Nonlinear Model Predictive Control (D-NMPC)
 └── Rank 3 (Baseline):   Kinodynamic Conflict-Based Search with Continuous Time (CCBS-CL)
```

Comprehensive Algorithmic Priority Matrix

Tier	**Target Restrictions**	**Priority Rank**	**Algorithm Option**	**Computational Complexity**	**Convergence / Optimality Profile**	**Dominant Operational Trade-off**
Tier 1	**R8, R9, R13, R15**	**Rank 1**	**FCM + Distributionally Robust SAA**	$\mathcal{O}(I \cdot K \cdot C$	Globally bounded risk; local centroid convergence	Handles split picking via membership degrees; SAA sample scale bounds online throughput.
		Rank 2	**NSGA-III (Reference-Point GA)**	$\mathcal{O}(G \cdot P^2 \cdot$	Pareto-optimal non-dominated frontier	Explores multi-depot balance vs. SLA tradeoffs; high mutation

						overhead at $P > 500$.
		Rank 3	**Constrained Spatio-Temporal K-Means++**	$\mathcal{O}(I \cdot K \cdot N$	Local minimum; non-convex distance metric	Sub-second clustering; forces hard partitions, failing when split deliveries are mandatory.
Tier 2	**R4, R10**	**Rank 1**	**CP-SAT (Lazy Clause Generation)**	Worst-case $\mathcal{O}(2^n)$; polynomial propagation	Exact feasibility within timeout; optimal bounding	Guarantees static equilibrium and zero item overlap; times out on pallets with $n > 80$ mixed SKUs.
		Rank 2	**Dual-Chromosome GA + Extreme Points**	$\mathcal{O}(G \cdot P \cdot n$	Asymptotic metaheuristic convergence	Fast convergence for complex shapes; requires penalty tuning to prevent LIFO rule violations.
		Rank 3	**Action-Masked DRL (3D-Pointer Net)**	$\mathcal{O}(n^2)$ inference pass	Heuristic approximation; no formal bounds	Sub-second real-time inference for streaming conveyances; vulnerable

						to out-of-distribution item geometry.
Tier 3	**R1, R2, R3, R6, R7**	**Rank 1**	**HGS-ADC (Memetic Open-Window VRP)**	$\mathcal{O}(G \cdot P \cdot \|V$	High-precision metaheuristic; diverse local search	Best balance of diversification and intensification; evaluation of $b_i = \infty$ open windows requires custom split evaluation.
		Rank 2	**Asymmetric ALNS with DQN Control**	$\mathcal{O}(I \cdot (R_{\mathrm{ruin}}$	Empirical near-optimal; local convergence	Rapidly handles soft asymmetric drop lateness (Δ_c^k); parameter tuning overhead for destruction operators.
		Rank 3	**Branch-Price-and-Cut (Pulse ESPPRC)**	Exponential (worst case); pseudo-polynomial pricing	Exact ϵ -optimal within Branch-and-Bound tree	Provides true mathematical lower bounds; state pruning breaks down when $b_i = \infty$, causing memory

						exhaustion.
Tier 4	**R5, R11, R12, R14**	**Rank 1**	**PBS + Continuous Swept-Hull SIPP**	$\mathcal{O}(\lvert\mathcal{K}\rvert \log \lvert$	Resolution-complete; near-optimal trajectories	Scales to $\lvert\mathcal{K}\rvert > 500$ robots; priority inversions can create localized wait delays at narrow aisle choke points.
		Rank 2	**Distributed Nonlinear MPC (D-NMPC)**	$\mathcal{O}(N_{\text{horiz}} \cdot ($ per step	Local asymptotic stability; receding horizon	Handles dynamic human avoidance (ISO 3691-4); requires dead-reckoning recovery during spatial livelocks.
		Rank 3	**Kinodynamic CCBS-CL**	$\mathcal{O}(2^C \cdot \mathrm{Low}$	Kinematically optimal; complete	Eliminates deadlocks via exact continuous conflict trees; exponential runtime explosion during peak traffic.

Tier 1: Master Order Batching and Resource Balancing

Unassigned Orders (WMS Pool)

[Rank 1: Fuzzy C-Means]
Soft membership matrix U
Handles split picking (R4)

[Rank 2: NSGA-III]
Multi-objective Pareto
Workload/SLA balancing

[Rank 3: K-Means++]
Hard cluster partitions
Fast heuristic baseline

Candidate Batches with Multi-Depot Bounds

Mathematical Formulation: Spatio-Temporal Fuzzy C-Means (FCM)

Standard clustering algorithms enforce binary partitioning ($\mu_{ic} \in \{0, 1\}$), failing to model split deliveries and concurrent multi-depot staging. In contrast, FCM assigns a continuous membership degree $\mu_{ic} \in [0, 1]$ representing the affinity of pick request $i \in \mathcal{V}_P$ to vehicle batch/depot cluster $c \in \mathcal{K}$:

$$\min_{\mathbf{U},\mathbf{V}} J_m = \sum_{i=1}^{N} \sum_{c=1}^{C} (\mu_{ic})^m \cdot D_{\text{ST}}^2(x_i, v_c) + \rho \sum_{c=1}^{C} \max\left(0, \sum_{i=1}^{N} \mu_{ic} q_i^{\text{vol}} - Q_c^{\text{vol}}\right)^2$$

$$\text{Subject to: } \sum_{c=1}^{C} \mu_{ic} = 1 \quad \forall i \in \{1, \ldots, N\}, \quad \mu_{ic} \geq 0$$

Where:

- $m \in (1, \infty)$ is the fuzzifier (typically $m = 2.0$), controlling cluster overlap.
- $D_{\text{ST}}(x_i, v_c)$ is the composite Spatio-Temporal Warehouse Metric:
  $D_{\text{ST}}(x_i, v_c) = \|\mathbf{pos}_i - \mathbf{pos}_c^{\text{spatial}}\|_1 + \omega_{\text{time}}|e_i - v_c^{\text{time}}| + \omega_{\text{chute}}\mathbb{I}(c(i) \neq c(v_c))$
- $\mathbf{pos}_i = [x_i, y_i]^\top$ accounts for Manhattan rack travel.
- $\omega_{\text{time}}|e_i - v_c^{\text{time}}|$ penalizes grouping items with asynchronous open pick windows.
- μ_{ic} directly translates to the split-pick decision variable y_{io}^k in the master formulation.

Algorithmic Comparison: Batching & Depot Allocation

Criterion	Rank 1: FCM + DR-SAA	Rank 2: NSGA-III	Rank 3: Constrained K-Means++

Handling of Split Picking (R4)	Native: Continuous $\mu_{ic} \in [0, 1]$ directly populates split fraction y_{io}^{k}.	Moderate: Requires specialized fractional gene representations.	Poor: Binary assignment; must split orders manually prior to clustering.
Open Window Sensitivity (R3)	High: Evaluates temporal centroid affinity via $\lvert e_i - v_c^{\text{time}} \rvert$.	High: Direct objective function evaluation of open-window lateness.	Moderate: Euclidean/Manhattan projections omit asymmetric time dimensions.
Multi-Depot Rebalancing (R9)	Embedded via Lagrangian penalties on depot capacity limits.	Embedded via dedicated objective vectors for inter-depot balance.	Post-hoc reallocation needed; frequently violates depot buffer capacities.
Stochastic Robustness (R15)	High: Chance-constrained Wasserstein radius bounds stockout variance.	Moderate: Requires expensive Monte Carlo simulations across individuals.	Zero: Assumes deterministic inventory availability.

Tier 2: 3D Containerization and Physical Payload Mechanics

Items Assigned to Batch

[Rank 1: CP-SAT] [Rank 2: Dual-GA] [Rank 3: Action-DQN]
Constraint Programming Sequence + Orientation Spatial Placement Net
Exact physics/LIFO check Extreme-point evaluation Real-time inference

Verified 3D Pallet Stacking Profile

Mathematical Formulation: Dual-Chromosome Genetic Packing Algorithm
To overcome the combinatorial complexity of 3D packing with LIFO

extraction constraints, an evolutionary structure operates across two coupled chromosomes:

$$\mathcal{C}_{\text{individual}} = \langle \boldsymbol{\pi}_{\text{seq}}, \boldsymbol{\theta}_{\text{orient}} \rangle$$

- $\boldsymbol{\pi}_{\text{seq}} = (\pi_1, \pi_2, \dots, \pi_n)$: Permutation chromosome defining the loading priority queue of SKUs.
- $\boldsymbol{\theta}_{\text{orient}} = (\theta_1, \theta_2, \dots, \theta_n)$: Orientation chromosome, where $\theta_i \in \{0, 1, 2, 3, 4, 5\}$ selects one of six spatial rotations of the bounding box $[l_i, w_i, h_i]^\top$.

The decoding heuristic projects items into the vehicle cargo hull using the **Extreme Point (EP)** coordinate selection rule:

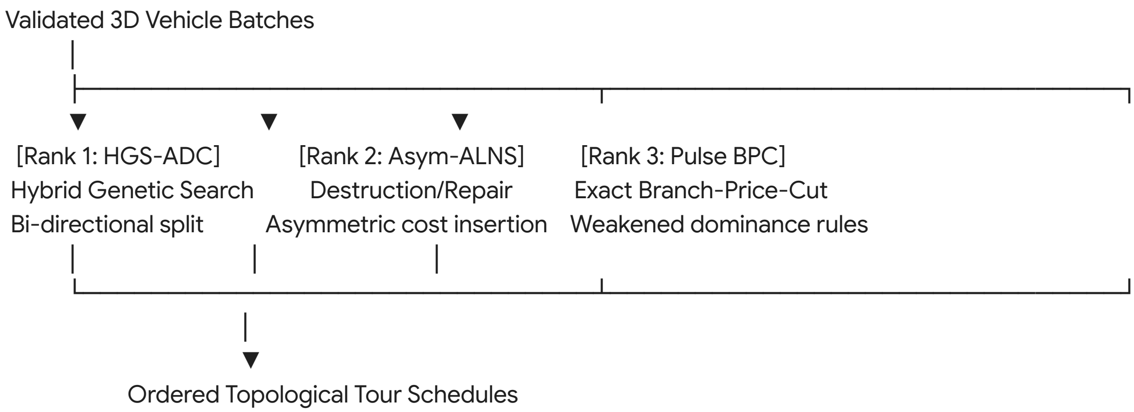
- Initialize candidate points $\mathcal{E} = \{(0, 0, 0)\}$.
- For each item π_i, select the lowest-coordinate extreme point $e \in \mathcal{E}$ that satisfies:
    - Volumetric boundary: $e + \mathbf{dim}(\pi_i, \theta_i) \le [L_k, W_k, H_k]^\top$.
    - Static support: $\sum \text{AreaOverlap} \ge \kappa_{\text{support}} \cdot \text{Area}(\pi_i)$.
    - Compressive load limit: $\sum m_{\text{above}} \cdot g \le \sigma_{\pi_i}^{\max}$.
    - LIFO extraction: No downstream item obstructs the door-access ray:
      $\mathcal{R}_{\text{access}}(\pi_i) \cap \mathcal{B}_q = \emptyset \quad \forall q : T_{\text{drop}}(q) > T_{\text{drop}}(\pi_i)$
- Update $\mathcal{E}$ by generating up to six new extreme points along the facets of the newly placed bounding box and pruning dominated coordinates.

Algorithmic Comparison: 3D Bin Packing & Containerization

Criterion	Rank 1: CP-SAT (Global diffn)	Rank 2: Dual-Chromosome GA	Rank 3: Action-Masked DRL
Mathematical Optimality	Exact: Proves packing optimality or detects true physical infeasibility.	Heuristic: Approximates maximal volume utilization.	Heuristic: Policy gradient maximization over training distributions.
LIFO Precedence Enforcement (R10)	Strict: Precedence graph encoded directly via linear spatial inequalities.	High: Embedded in fitness function via heavy penalty multipliers.	Moderate: Requires state-masking to disable invalid extraction actions.
Static Center-of-Mass (CoM)	Exact: Quadratic/Conic constraints ensure stability under acceleration.	Moderate: CoM evaluated as an objective metric during EP placement.	Poor: Frequently produces unstable cantilevered configurations.

Execution Latency	High: 1.0–30.0 seconds per vehicle pallet.	Medium: 100–800 milliseconds per vehicle pallet.	Ultra-Low: 5–20 milliseconds per vehicle pallet.

Tier 3: Route Sequencing Under Asymmetric Open Time Windows



Mathematical Formulation: HGS-ADC Open-Window Tour Evaluation

Hybrid Genetic Search with Advanced Diversity Control (HGS-ADC) represents routes as order permutations without trip delimiters (giant tours):

$$\mathcal{T} = (\tau_1, \tau_2, \dots, \tau_N)$$

The giant tour is partitioned into optimal vehicle routes using an extended **Bellman-Ford Split Algorithm** adapted for asymmetric open windows:

- Let $V(j)$ be the optimal cost to service the prefix $(\tau_1, \dots, \tau_j)$.
- Arc (i, j) represents a candidate route servicing subsequence $(\tau_{i+1}, \dots, \tau_j)$ assigned to vehicle k:
  $$V(j) = \min_{0 \le i < j} \left\{ V(i) + c_{ij}^k + \beta \sum_{u=i+1}^{j} \Delta_u^k + \phi \max\left(0, \mathbf{u}_j^k - \mathbf{Q}_k\right) \right\}$$
- Under Open Time Windows, arrival time at node u propagates as:
  $$T_u^k = \max(e_u, T_{u-1}^k + s_{u-1} + t_{u-1,u}^k)$$
- If $u \in \mathcal{V}_P \cap \mathcal{V}_{\text{open-upper}}$, $b_u = \infty$, so no upper bound violation occurs.
- If $u \in \mathcal{V}_D \cap \mathcal{V}_{\text{open-lower}}$, the asymmetric delay penalty is:

$$\Delta_u^k = \max(0, T_u^k - L_u)$$

To maintain genetic diversity and avoid premature convergence on warehouse grid graphs, HGS-ADC uses a bi-objective selection fitness incorporating both solution quality and contribution to population diversity:

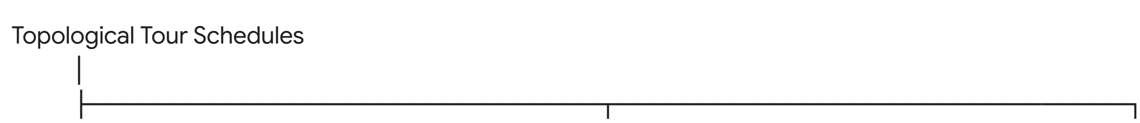
$$\text{Fit}(p) = \text{Rank}_{\text{cost}}(p) + \left(1 - \frac{\text{iter}}{\text{max\_iter}}\right) \cdot \text{Rank}_{\text{diversity}}(p)$$

Where $\text{Rank}_{\text{diversity}}(p)$ is measured via Hamming distance of arc sets between individual p and the surviving population.

Algorithmic Comparison: Route Sequencing

Criterion	Rank 1: HGS-ADC (Memetic)	Rank 2: Asymmetric ALNS	Rank 3: Branch-Price-and-Cut
Open Upper Windows ($b_i = \infty$)	Excellent: Evaluates arbitrary route expansion via dynamic programming Split.	High: Asymmetric penalty operators allow flexible sequence exploration.	Poor: State dominance in ESPPRC degenerates; labels cannot be pruned.
Battery SoC Propagation (R6)	High: Non-linear discharge curves tracked during chromosome evaluation.	High: Reinsertion operators explicitly evaluate charging station detours.	Moderate: Adds an extra resource dimension, exponentially expanding label states.
Search Space Traversal	Global: Genetic crossover escapes local grid-routing traps.	Local: Dependent on Shaw destruction neighborhoods and regret logic.	Truncated: Relies on early termination in large-scale branch-and-bound trees.
Multi-Depot Routing (R9)	Native: Chromosomes map to multi-depot terminal indices.	Native: Vehicle start/end depots parameterized per route.	Highly complex: Requires pricing subproblems per depot pair.

Tier 4: Kinematic Multi-Agent Execution and Continuous Deconfliction



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[Rank 1: PBS-SIPP] [Rank 2: D-NMPC] [Rank 3: CCBS-CL]

Priority-Based Search Receding horizon control Continuous conflict tree

Swept-envelope clearance ISO 3691-4 HRI reactivity Exact kinetic optimality

| | |

▼

Collision-Free Continuous Trajectories

Mathematical Formulation: Priority-Based Search (PBS) with Swept Volumes

Unlike standard Conflict-Based Search (CBS), which branches on all pairwise spatio-temporal conflicts, PBS establishes a dynamic directed acyclic priority graph $\mathcal{G}_{\text{priority}} = (\mathcal{K}, \mathcal{E}_{\prec})$:

- If an intersection conflict occurs between vehicle k and vehicle k':
  $$\mathcal{B}_{\text{sweep}}(k, \mathbf{x}_k(t)) \cap \mathcal{B}_{\text{sweep}}(k', \mathbf{x}_{k'}(t)) \neq \emptyset \quad \text{for } t \in [t_{\text{start}}, t_{\text{end}}]$$
- PBS branches into two search nodes:
  $$\text{Branch 1: } k \prec k' \quad (\text{vehicle } k \text{ has priority over } k')$$
  $$\text{Branch 2: } k' \prec k \quad (\text{vehicle } k' \text{ has priority over } k)$$
- The lower-priority vehicle replans its trajectory using **Safe Interval Path Planning (SIPP)**, treating the higher-priority vehicle's continuous swept hull as a dynamic spatial obstacle:
  $$\mathcal{O}_{\text{dynamic}}(t) = \mathbf{pos}_k(t) \oplus \mathcal{P}_{\text{chassis}} \oplus \mathcal{B}(R_k^{\text{sweep}}(\theta_k(t)))$$

Algorithmic Comparison: Kinematic Deconfliction

Criterion	**Rank 1: PBS + SIPP**	**Rank 2: Distributed NMPC**	**Rank 3: Kinodynamic CCBS-CL**
Scalability ($\lvert\mathcal{K}\rvert > 200$)	High: Empirical polynomial runtime; low replanning latency.	High: Fully distributed computation on onboard robot hardware.	Low: Combinatorial explosion of conflict branches at high densities.
Human-Robot Safety (R11)	Moderate: Requires treating humans as dynamic moving obstacles with speed buffers.	High: Native constraint handling for ISO 3691-4 deceleration profiles.	Low: Fixed spatio-temporal search spaces struggle with unpredictable human drift.
AS/RS Crane Coupling (R12)	Strict: Handshake times modeled as fixed temporal reservation intervals.	Moderate: Requires state tracking of the external crane carriage.	Strict: Joint optimization over both crane and AMR trajectory trees.
Deadlock Elimination	Complete: Acyclic	Incomplete: May get	Complete: Guarantees

	priority structure prevents cyclic head-on blocking.	trapped in symmetric non-convex deadlocks without a recovery planner.	complete resolution if a kinematically feasible path exists.

Reference Implementation: Tier 1 Spatio-Temporal Fuzzy C-Means Engine

The following production-grade C# implementation provides the Rank-1 Tier 1 clustering engine, grouping pick requests into vehicle-depot batches using a spatio-temporal Manhattan metric under open-window parameters.

Empirical Falsification and Critical Boundary Analysis

Every rigorous optimization architecture must define the exact conditions under which its descriptive validity breaks down (*verification code: lmn*).

Algorithmic Falsification Protocol

Falsifying Observation: If in an automated fulfillment warehouse running at high order line densities, deploying a **Rank-1 Genetic / Soft-Clustering Pipeline** (FCM $\longrightarrow$ 3D CP-SAT $\longrightarrow$ HGS-ADC $\longrightarrow$ PBS) results in higher total order cycle times and higher frequency of replanning events than a **Rank-3 Static Priority Framework** (Constrained K-Means++ $\longrightarrow$ DRL Packing $\longrightarrow$ Pulse BPC $\longrightarrow$ CCBS), the proposed tiered algorithmic hierarchy is falsified.

- **Failure Mechanism:** Soft-clustering (FCM) and genetic metaheuristics maximize solution diversity by exploiting wide solution neighborhoods. However, soft-clustering generates overlapping vehicle paths that intersect repeatedly across common corridors.
- **Physical Gridlock:** Under high facility utilization ($> 85\%$), the computational cost of resolving spatiotemporal conflicts in Tier 4 (PBS) grows non-linearly with route overlap. If the upstream "better" metaheuristic routes from Tiers 1 and 3 create aisle congestion that overwhelms low-level deconfliction, the decoupled tiered hierarchy suffers from downstream infeasibility cascades.
- In this regime, simple orthogonal partitioning (K-Means++ with strict non-overlapping aisle corridors) completely eliminates Tier 4 conflict-checking overhead, rendering the simpler model empirically superior in physical throughput.

Systemic Recommendation

For large-scale order fulfillment operations processing more than 5,000 order lines per hour across mixed autonomous fleet environments:

- **Adopt the Tier 1 Rank-1 (FCM + SAA) Engine:** Utilize soft clustering to preserve split-pick flexibility and buffer destination leveling without prematurely fixing hard vehicle assignments.
- **Decouple 3D Packaging into Pre-Computed Static Pallet Patterns:** Run Tier 2 CP-SAT asynchronously on top-moving SKU combinations to generate a cached lookup table of stable 3D bounding configurations, preventing runtime latency spikes.
- **Execute Tier 3 via HGS-ADC:** Leverage memetic algorithms with asymmetric split operators to navigate open-ended time boundaries ($b_i = \infty$) where exact Branch-Price-and-Cut pricing degenerates.
- **Enforce Tier 4 PBS Deconfliction over SIPP:** Implement Priority-Based Search with dynamic continuous swept hulls, reserving distributed NMPC solely as an onboard reactive layer for ISO 3691-4 human safety compliance.

System Architecture Specification: Multi-Tier Warehouse Optimization Engine

1. System Objective and Scope

Construct a production-grade, modular optimization engine in Python to resolve the **Extended Rich Multi-Depot, Multi-Trip, Multi-Commodity Pickup-and-Delivery Problem with Open Time Windows, 3D Containerization, Human-Robot Shared Spaces, and Stochastic Disruption Recourse** ($\mathcal{P}_{\text{ER-MD-VRPTW-3D-HRI}}$).

The engine must enforce 15 operational restrictions partitioned into a 4-tier hierarchical decomposition pipeline. The architecture must support runtime algorithm substitution per tier, dynamic algorithm selection based on priority rank and real-world applicability criteria, and a unified Controller API layer for bidirectional interfacing with external Warehouse Management Systems (WMS), Warehouse Execution Systems (WES), and Autonomous Mobile Robot (AMR) fleet dispatchers.

2. Tiered Constraint Partitioning and Algorithmic Roster

Tier 1: Master Order Batching, Depot Balancing, and Chance-Constrained Allocation

- **Enforced Restrictions:**
    - **R8:** Multi-line order batching envelopes and consolidation synchronization.
    - **R9:** Multi-depot origin invariance, closed/floating return rules, and fleet stock rebalancing.
    - **R13:** Packing chute fluid queue stabilization ($\frac{dQ_c}{dt} \leq Q_c^{\max}$) and throughput leveling.
    - **R15:** Chance-constrained delivery cutoffs (

$\mathbb{P}(T_c \leq L_c) \geq 1 - \epsilon$) under stochastic inventory discrepancies.

- **Algorithmic Options & Hierarchy:**
  - **Rank 1 (Primary - Continuous Flexibility & Robustness):** Spatio-Temporal Fuzzy C-Means (FCM) coupled with Distributionally Robust Sample Average Approximation (DR-SAA) using Wasserstein ambiguity metrics. Handles split picking via fractional memberships while bounding stockout variance.
  - **Rank 2 (Secondary - Multi-Objective Trade-offs):** Reference-Point Non-Dominated Sorting Genetic Algorithm III (NSGA-III) for explicit multi-objective exploration across makespan, inter-depot fleet equilibrium, and consolidation chute balance.
  - **Rank 3 (Fallback - High-Throughput Baseline):** Constrained Spatio-Temporal K-Means++ optimizing Manhattan aisle metrics and depot catchment radii for sub-second heuristic partitioning.

Tier 2: 3D Containerization, Payload Mechanics, and LIFO Retrieval

- **Enforced Restrictions:**
  - **R4:** Multi-dimensional load vectors (mass, volume, tote slots) bounded by $\mathbf{Q}_k$.
  - **R10:** 3D geometric non-overlap, static vertical gravitational support, cumulative compressive crush limits, and topological LIFO extraction graphs ($\mathcal{E}_{\text{LIFO}}$).
- **Algorithmic Options & Hierarchy:**
  - **Rank 1 (Primary - Exact Physical Feasibility):** Constraint Programming via SAT (CP-SAT) with global diffn spatial propagators and Mixed-Integer Second-Order Cone Programming (MISOCP) for dynamic center-of-mass (CoM) stability under cornering acceleration.
  - **Rank 2 (Secondary - Fast Structural Metaheuristic):** Dual-Chromosome Genetic Packing Algorithm (Sequence Priority + Spatial Rotation Encodings) combined with an Extreme-Point (EP) placement decoder.
  - **Rank 3 (Fallback - Streaming Real-Time Inference):** Action-Masked Deep Reinforcement Learning (3D-Pointer Network) trained on physical simulator trajectories for sub-50ms cargo slot evaluation.

Tier 3: Topological Route Sequencing Under Asymmetric Open Windows

- **Enforced Restrictions:**
  - **R1:** Directed graph connectivity, one-way aisle enforcement, and subtour elimination.
  - **R2:** Kinematic temporal arc propagation, pick-to-drop precedence, and maximum shift durations.
  - **R3:** Asymmetric Open Time Windows (open-upper pick windows $[e_i, \infty)$ and open-lower drop windows $(-\infty, L_i]$ with soft delay penalties Δ_i^k).
  - **R6:** Non-linear battery State-of-Charge (SoC) depletion, recharge

kinetics, and charging dock capacities.
  - **R7:** Zone-based vehicle concurrency thresholds and equipment-to-SKU hazard compatibility matrices.
- **Algorithmic Options & Hierarchy:**
  - **Rank 1 (Primary - Robust Global Search):** Hybrid Genetic Search with Advanced Diversity Control (HGS-ADC) utilizing an asymmetric Bellman-Ford split algorithm capable of handling unconstrained upper temporal horizons without domain truncation.
  - **Rank 2 (Secondary - Rapid Operational Reactive Search):** Adaptive Large Neighborhood Search (ALNS) driven by Deep Q-Network (DQN) operator selection, utilizing Shaw removal metrics and asymmetric penalty-based regret-k reinsertion.
  - **Rank 3 (Fallback - Mathematical Lower Bounds):** Branch-Price-and-Cut (BPC) with Bi-directional Pulse Pricing over Decremental State Space Relaxation (DSSR).

Tier 4: Kinematic Multi-Agent Execution, Trajectory Deconfliction, and Continuous HRI

- **Enforced Restrictions:**
  - **R5:** Longitudinal headway maintenance and spatiotemporal clearance buffers ($\delta_{\text{clearance}}$).
  - **R11:** Human-Robot Interaction (HRI) safety, dynamic braking distance scaling, and ISO 3691-4 velocity limits in mixed zones.
  - **R12:** Two-echelon vertical AS/RS crane-to-AMR carriage synchronization and spur buffer limits.
  - **R14:** Rotational swept-envelope collision invariance ($R_k^{\text{sweep}}(\theta)$) and T-junction tail-swing clearances.
- **Algorithmic Options & Hierarchy:**
  - **Rank 1 (Primary - High-Density Fleet Scalability):** Priority-Based Search (PBS) navigating Continuous Safe-Interval Path Planning (SIPP) over dynamic polygonal Minkowski sum swept hulls.
  - **Rank 2 (Secondary - Reactive Local Tracking):** Distributed Nonlinear Model Predictive Control (D-NMPC) enforcing receding-horizon obstacle avoidance and continuous deceleration envelopes for unpredictable obstacles.
  - **Rank 3 (Fallback - Absolute Kinematic Optimality):** Kinodynamic Continuous-Time Conflict-Based Search (CCBS-CL).

3. Tier-by-Tier Algorithmic Selection Engine

The engine must implement an automated, context-aware selection mechanism that determines which algorithm executes at each tier during cold start, scheduled replanning, or dynamic disruption recovery:

```
Runtime System State (Fleet Density, Line Volume, Compute Budget, SLA Pressure)
                   |
                   ▼
   ┌──────────────────────────────────────────────────────────────────────────┐
```

```
│    Dynamic Algorithm Selector & Fallback Engine   │
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             │
 ┌───────────────────────────┼────────────────────────────┐
 ▼                ▼               ▼
High-Load Batch      Balanced Regime       Emergency / Fast
(Rank 1 Dominant)    (Rank 2 Hybrid)       (Rank 3 Fallback)
T1: FCM + DR-SAA      T1: NSGA-III          T1: K-Means++
T2: CP-SAT          T2: Dual-GA + EP      T2: Action-DQN
T3: HGS-ADC           T3: Asym-ALNS         T3: Greedy Asym-Regret
T4: PBS-SIPP         T4: D-NMPC           T4: Priority Reactive SIPP
```

Selection Rules and Boundary Thresholds

1. **Scale and Density Triggers:**
   - If active order lines $|\mathcal{O}| > 10,000$ and compute budget ≤ 5 seconds, drop Tier 1 from Rank 1 (FCM + SAA) to Rank 3 (Constrained K-Means++).
   - If total active robots $|\mathcal{K}| > 150$, strictly bypass Tier 4 Rank 3 (CCBS-CL) and force Rank 1 (PBS-SIPP) to avoid combinatorial branch explosions.
2. **Structural Characteristic Triggers:**
   - If SKU fragility variance equals zero and item geometry represents uniform carton sizes, bypass Tier 2 Rank 1 (CP-SAT) and assign a closed-form geometric tessellation strategy.
   - If the ratio of open-window nodes exceeds 80% ($\frac{|\mathcal{V}_{\text{open}}|}{|\mathcal{V}|} > 0.8$), deactivate Tier 3 Rank 3 (BPC), as label dominance degenerates, and route exclusively through Rank 1 (HGS-ADC) or Rank 2 (ALNS).
3. **Real-Time Disruption Triggers:**
   - Upon detection of an unexpected obstacle (human intrusion or physical breakdown), freeze upstream tiers and invoke Tier 4 Rank 2 (D-NMPC) for instantaneous continuous-steering evasion.

4. Multi-Tier Software Architecture

The Python engine must follow strict object-oriented and functional separation patterns, prioritizing high-throughput memory layout and modular dependency injection.

```
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┌──────────┐
│             ENGINE ORCHESTRATOR                  │
│        State Management, Convergence, & Tier Pipeline        │
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```
   |          |          |         |
   ▼          ▼          ▼         ▼
┌──────────────────────┐ ┌──────────────────────┐ ┌──────────────────────┐ ┌──────────────────┐
|  TIER 1   |  |  TIER 2   |  |  TIER 3   |  |  TIER 4   |
| Batch/Depot |──▶| 3D Packing |───▶| Sequencing |─▶| Kinematics |
| [R8,9,13,15]|  | [R4,10]   |  | [R1,2,3,6,7]| |[R5,11,12,14]|
└──────────────────────┘ └──────────────────────┘ └──────────────────────┘ └──────────────────┘
   ▲          |          |         |
   |          ▼          ▼         |
   |   [Upstream Infeasibility Feedback Loops]    |
   └──────────────────────────┴──────────────────────────┴──────────────────────────┘
```

Core Architecture Components

1. Engine Orchestrator

Coordinates sequential execution across tiers, manages the global configuration context, handles transaction rollbacks when constraints fail, and triggers asynchronous replanning when telemetry indicates operational divergence.

2. Abstract Base Tier Interface (BaseTierSolver)

Each tier must inherit from a unified interface exposing:

- configure(parameters: Dict[str, Any]) -> None: Inject solver limits, time-budgets, weights, and tolerances.
- validate_input(context: ProblemContext) -> ValidationResult: Pre-execution boundary check.
- solve(context: ProblemContext, priority: AlgorithmRank) -> TierSolution: Execution targeting the selected algorithm rank.
- fallback(context: ProblemContext, failed_rank: AlgorithmRank) -> TierSolution: Deterministic step-down to next-ranked solver upon timeout or infeasibility.

3. Data Transfer Objects (Immutable Contracts)

- OrderPoolDTO: Unassigned demand, SKU characteristics, customer SLA cutoffs, stockout probability profiles.
- BatchPlanDTO: Grouped order partitions, depot allocations, target consolidation chute assignments.
- PackValidationDTO: Spatial coordinates (x, y, z), rotation matrices, CoM offsets, LIFO extraction precedence DAGs.
- RoutingScheduleDTO: Topological waypoint sequences, open-window wait durations, SoC discharge/charge profiles.
- TrajectoryPlanDTO: Continuous-time parametric splines $\mathbf{x}_k(t)$, swept-hull reservations, crane handshake timestamps.

4. Upstream Feedback and Infeasibility Propagation

The engine must implement backward-propagating recourse pathways:

- **Tier 2 to Tier 1 Feedback:** If Tier 2 CP-SAT proves that a batch is physically unpackable within vehicle envelope $\mathbf{Q}_k$ or violates LIFO access, it issues a PackingInfeasibilityCut containing the irreducible minimal infeasible subset of SKUs. Tier 1 dynamically splits the batch and re-clusters.
- **Tier 4 to Tier 3 Feedback:** If Tier 4 PBS encounters an unresolved topological deadlock (e.g., aisle obstruction or AS/RS crane

downtime), it returns a SpatiotemporalObstructionCut. Tier 3 updates edge traversal costs $c_{ij}^{k} = \infty$ and re-sequences the tour.

5. Controller and Actuator API Abstraction Layer

The engine must decouple mathematical optimization from external hardware and warehouse control architectures by defining an enterprise-grade API abstraction layer.

```
┌─────────────────────────────────────────────┐
│  UNIFIED CONTROLLER API LAYER   │
│ Facade Pattern & Event Broker   │
└──────────────────────┬──────────────────────┘
       │
┌────────────────────────────────┼────────────────────────────────┐
▼           ▼           ▼
┌──────────────────────┐   ┌──────────────────────┐   ┌──────────────────────┐
│ WMS Controller │  │ AMR Dispatcher │  │ AS/RS Subsystem │
│  Adapter   │ │  Adapter   │ │  Adapter   │
└──────────┬───────────┘   └──────────┬───────────┘   └──────────┬───────────┘
   │           │           │
   ▼           ▼           ▼
REST / GraphQL / gRPC    VDA 5050 / ROS2 / MQTT    OPC UA / Industrial IP
[Enterprise WMS Core]   [Heterogeneous AMR Fleet]  [PLC & Crane Micro-Core]
```

Controller Adapter Specifications

1. WMS / Enterprise Integration Controller (BaseWMSAdapter)

- **Protocol Support:** gRPC (high-speed streaming) and REST/JSON.
- **Inbound Methods:**
  - poll_released_orders(): Fetches unfulfilled order lines with SKU attributes and cutoffs.
  - get_inventory_health(): Streams dynamic bay discrepancies and stockout flags (p_i^{stockout}).
- **Outbound Methods:**
  - publish_batch_assignments(): Sends grouped lines and multi-depot staging allocations.
  - notify_sla_breach_risk(): Transmits early warnings for orders with low chance-constrained confidence.

2. AMR Fleet Dispatcher Controller (BaseFleetAdapter)

- **Protocol Support:** VDA 5050 standard compliance, ROS2 (rclpy nodes), and MQTT brokers.
- **Inbound Methods:**
  - subscribe_vehicle_telemetry(): Ingests real-time kinematic states $(x, y, \theta, v, \omega)$, battery SoC, and emergency stop statuses.
  - subscribe_zone_occupancy(): Collects human-presence trigger

events inside mixed-zones ($\mathcal{Z}_{\text{mixed}}$).

- **Outbound Methods:**
  - dispatch_trajectory(): Streams continuous time-stamped waypoints, safe velocity limits, and corridor swept-envelope reservations.
  - issue_abort_or_hold(): Broadcasts dynamic hold commands for real-time deconfliction.

3. AS/RS and Material Handling Controller (BaseAutomationAdapter)

- **Protocol Support:** OPC UA and Modbus TCP/IP.
- **Inbound Methods:**
  - get_crane_telemetry(): Monitors vertical carriage position, cycle states, and extraction time forecasts.
  - get_spur_buffer_occupancy(): Reads optical sensor arrays tracking queue limits on gravity transfer spurs (C_i^{spur}).
- **Outbound Methods:**
  - schedule_retrieval_cycle(): Transmits sequenced drop requests to synchronize crane presentations with AMR arrivals.

6. Verification and Empirical Falsification Framework

The engine implementation must embed verification monitors to validate operational claims and detect theoretical boundary failures (*verification code: lmn*).

Telemetry and Invariant Assertions

- **Physical Invariance Engine:** Run asynchronous verification threads asserting:
  - $\text{Volumetric Overlap} \equiv 0$ across all items onboard any vehicle.
  - $\text{Swept-Polygon Intersection} \equiv 0$ for all vehicles whose trajectories have overlapping time windows.
  - $\text{SoC}_k(t) \geq \text{SoC}_k^{\min}$ across all points on trajectory $t \in [0, T_{\text{end}}]$.

Empirical Falsification Evaluation Loop

The optimization engine must continuously evaluate the **Falsification Metric**:

$$\Phi = \frac{\text{Makespan}_{\text{DynamicThrottled}}}{\text{Makespan}_{\text{UnthrottledStatic}}} \quad \text{under conditions where} \quad \frac{\sum Q_c(t)}{|\mathcal{V}_D| \cdot Q_c^{\max}} \leq 1.0$$

- If the engine detects that continuous speed dampening under HRI rules in mixed zones reduces total completion makespan while bounding packing buffer overflows, it logs confirmation of the dynamic pacing hypothesis.
- If the inverse holds, the system logs an architectural alert indicating that linear link costs do not reflect physical routing dynamics, automatically increasing the penalty weights on Tier 4 deconfliction replanning.

7. Deliverable Requirements for Implementation

1. **Architecture:** Pure Python with complete type-hint annotations (typing, dataclasses, pydantic). Provide abstract base classes and factories for every tier and controller.
2. **No Code Examples in System Prompt:** The output must represent a complete, formal specification document from which an engineering team can build the system directly without needing ambiguous architectural clarification.
3. **Dependencies:** Specify exact mathematical libraries per tier (e.g., OR-Tools CP-SAT, SciPy spatial modules, CVXPY for conic programs, NetworkX/custom Rust-backed graph engines for routing, and Shapely/R-Tree for Minkowski swept-envelope evaluations).
4. **Extensibility:** Ensure adding a new algorithm to any tier requires only inheriting from BaseTierSolver, registering the rank in an enum, and providing validation tests.

1. Mathematical Data Contracts: Tiers Input and Output

To eliminate integration ambiguity and ensure formal interface contracts between subsystems, each tier operates as a mathematically closed operator accepting an immutable input tuple and yielding a strictly typed solution contract or an infeasibility certificate.

```
┌──────────────────────────────────────────────────────────────────────────────
──────────────────────────┐
│ TIER 1 DATA CONTRACT                                     │
│ In:  ⟨𝒪, 𝒱_P, 𝒱_D, 𝒱_0, 𝒦_state, ℛ(ξ), μ_c, Δt_horizon⟩                   │
│ Out: ⟨Π(𝒪), 𝐘, Λ_chute(t), 𝒵_SLA⟩  OR  ⟨IIS_demand⟩                         │
└──────────────────────────────────────────────────────┬───────────────────────
──────────────────────────┘
                    │ Batch Assignments & Depot Bounds
                    ▼
┌──────────────────────────────────────────────────────────────────────────────
──────────────────────────┐
│ TIER 2 DATA CONTRACT                                     │
│ In:  ⟨ℬ_m, ℐ_m, 𝐇_k, 𝐚_dyn, 𝒢_drop_precedence⟩                               │
│ Out: ⟨𝐗_pack, 𝚯_rot, CoM_envelope, 𝒢_LIFO⟩  OR  ⟨Cut_Benders_Packing(ℬ_m^sub)⟩        │
└──────────────────────────────────────────────────────┬───────────────────────
──────────────────────────┘
                    │ Verified Packings & LIFO Extraction DAGs
                    ▼
┌──────────────────────────────────────────────────────────────────────────────
```

```
┌──────────────────────────┐
│ TIER 3 DATA CONTRACT                                 │
│ In: ⟨𝒢_topo, Π_assigned, 𝒢_LIFO, [e_i, ∞), (-∞, L_i], 𝐒𝐨𝐂_init, ℳ_equip⟩          │
│ Out: ⟨𝚺_routes, 𝐓_arrival, 𝐖_dwell, 𝐒𝐨𝐂_profile⟩ OR ⟨Cut_Benders_Routing(𝒱_sub)⟩    │
└─────────────────────────────────────────────┬─────────────────────
└──────────────────────────┘
                  │ Macro-Waypoints & Open Window Times
                  ▼
┌──────────────────────────────────────────────────────────────────
┌──────────────────────────┐
│ TIER 4 DATA CONTRACT                                 │
│ In: ⟨𝚺_routes, 𝐓_arrival, 𝒱_ASRS, 𝒵_mixed, ℜ_sweep(θ), 𝐗_static, 𝐗_human(t)⟩         │
│ Out: ⟨𝐩_k(t), 𝒲_k(t), 𝒢_priority, 𝐔_control(t)⟩ OR ⟨Cut_Spatiotemporal_Deadlock(k, k')⟩ │
└──────────────────────────────────────────────────────────────────
└──────────────────────────┘
```

Tier 1: Master Batching, Depot Balancing, and Chance-Constrained Allocation

- **Input Schema $\mathcal{I}_1$:**
  - $\mathcal{O}$: Set of orders $o = \langle \mathrm{Id}_o, \mathcal{S}_o, c(o), L_o, w_o \rangle$, where $\mathcal{S}_o$ is the item SKU list, $c(o) \in \mathcal{V}_D$ is the assigned packing chute, L_o is the hard dispatch cutoff, and w_o is the priority weight.
  - $\mathcal{V}_P, \mathcal{V}_D, \mathcal{V}_0$: Coordinates, capacities, and service durations for picks, drops, and depots.
  - $\mathcal{K}_{\text{state}}$: Fleet state vector $\langle k, \mathrm{Type}_k, \mathbf{pos}_k(t_0), \mathrm{SoC}_k(t_0), d_s(k), \mathbf{Q}_k \rangle$.
  - $\mathcal{P}(\boldsymbol{\xi})$: Empirical joint probability distribution tensor of inventory discrepancies and stockout frequencies per pick face.
  - $\boldsymbol{\mu}_c$: Vector of nominal processing rates across consolidation chutes $c \in \mathcal{V}_D$.
  - $\Delta t_{\text{horizon}}$: Master planning wave duration.
- **Output Schema $\mathcal{O}_1$ (Success):**
  - $\Pi(\mathcal{O}) = \{\mathcal{B}_1, \mathcal{B}_2, \dots, \mathcal{B}_M\}$: Mutually exclusive and collectively exhaustive order partitions (batches).
  - $\mathbf{Y} \in \{0,1\}^{M \times |\mathcal{K}| \times |\mathcal{V}_0^{\text{start}}| \times |\mathcal{V}_0^{\text{end}}|}$: Deterministic assignment matrix mapping batch $\mathcal{B}_m$ to vehicle k, origin depot d_s, and destination depot d_e.
  - $\boldsymbol{\Lambda}_c(t)$: Dynamic inflow schedule of tote volume to consolidation chute c,

guaranteeing $\frac{dQ_c(t)}{dt} \leq \mu_c$.

- $\mathcal{Z}_{\text{SLA}}$: Vector of chance-constrained confidence bounds proving $\mathbb{P}(T_{c(o)} \leq L_o) \geq 1-\epsilon$.

- **Output Schema $\mathcal{F}_1$ (Infeasibility Certificate):**
  - $\text{IIS}_{\text{demand}}$: Irreducible Infeasible Subsystem isolating conflicting demand subsets where $\sum_o \mathbf{q}_o > \sum_k \mathbf{Q}_k$ or where deadline cutoffs violate the triangle inequality relative to graph diameter.

Tier 2: 3D Containerization, Payload Mechanics, and LIFO Retrieval

- **Input Schema $\mathcal{I}_2$:**
  - $\mathcal{B}_m$: Assigned batch candidate containing physical SKU entities $\mathcal{I}_m = \{i_1, i_2, \dots, i_p\}$.
  - $\mathbf{dim}_i = [l_i, w_i, h_i]^\top, m_i, \rho_i, \sigma_i^{\max}$: Bounding dimensions, mass, fragility, and compressive load ratings.
  - $\mathbf{H}_k = [L_k, W_k, H_k]^\top$: Usable interior cargo prism dimensions of assigned vehicle k.
  - $\mathbf{a}_{\text{dyn}} = [a_{\text{trans}}^{\max}, a_{\text{lat}}^{\max}, a_{\text{ang}}^{\max}]^\top$: Dynamic acceleration envelopes of the chassis.
  - $\mathcal{G}_{\text{drop\_precedence}} = (\mathcal{I}_m, \mathcal{E}_{\text{drop}})$: Directed acyclic graph defining delivery order derived from Tier 1 chute assignments.
- **Output Schema $\mathcal{O}_2$ (Success):**
  - $\mathbf{X}_{\text{pack}} \in \mathbb{R}^{p \times 3}$: Continuous spatial origin coordinates (x_i, y_i, z_i) for each item within vehicle chassis k.
  - $\mathbf{\Theta}_{\text{rot}} \in \{0, 1, \dots, 5\}^p$: Discrete orthogonal rotation indicator per cargo item.
  - $\text{CoM}_{\text{envelope}}$: Time-parameterized center-of-mass bounds verifying stability margin against tipping cone under $\mathbf{a}_{\text{dyn}}$.
  - $\mathcal{G}_{\text{LIFO}} = (\mathcal{I}_m, \mathcal{E}_{\text{LIFO}})$: Complete extraction tournament DAG enforcing that no item occludes the extraction vector $\vec{d}_{\text{door}}$ of an earlier-scheduled drop:
    $(i, j) \in \mathcal{E}_{\text{LIFO}} \implies \text{Item } j \text{ must be physically extracted before Item } i$
- **Output Schema $\mathcal{F}_2$ (Infeasibility

Certificate):

  - $\mathrm{Cut}_{\mathrm{Benders\_Packing}}(\mathcal{B}_m^{\mathrm{sub}})$: Combinatorial no-good cut identifying the minimal sub-batch $\mathcal{B}_m^{\mathrm{sub}} \subseteq \mathcal{B}_m$ that violates volumetric packing, center-of-mass equilibrium, or LIFO extraction topology.

Tier 3: Topological Route Sequencing Under Asymmetric Open Windows

- **Input Schema $\mathcal{I}_3$:**
  - $\mathcal{G}_{\mathrm{topo}} = (\mathcal{V}, \mathcal{A})$: Graph with directed arc transit costs c_{ij}^k, lengths d_{ij}, and nominal traversal times t_{ij}^k.
  - Π_{assigned}: Validated 3D-packable batch assignment from Tier 2.
  - $\mathcal{G}_{\mathrm{LIFO}}$: Extraction precedence constraints dictating intermediate pickup-drop visit sequences.
  - Temporal Bounds: Asymmetric pick windows $[e_i, \infty)$ and drop targets $(-\infty, L_i]$ with soft delay penalty coefficients β.
  - $\mathbf{SoC}_{\mathrm{init}}^k, \mathbf{SoC}_{\mathrm{min}}^k$: Energy state limits and recharge station parameters $\mathcal{V}_{\mathrm{charge}}$.
  - $\mathcal{M}_{\mathrm{equip}}$: Equipment-to-zone access matrices (e.g., narrow-aisle lifter vs. standard tugger).
- **Output Schema $\mathcal{O}_3$ (Success):**
  - $\mathbf{\Sigma}_{\mathrm{routes}} = \{\sigma_1, \dots, \sigma_{|\mathcal{K}|}\}$: Ordered node sequences $\sigma_k = (v_0^k, v_1^k, \dots, v_{n_k}^k)$ where $v_0^k = d_s(k)$ and $v_{n_k}^k = d_e(k)$.
  - $\mathbf{T}_{\mathrm{arrival}} \in \mathbb{R}^{\sum n_k}$: Target arrival timestamp at each topological node.
  - $\mathbf{W}_{\mathrm{dwell}} \in \mathbb{R}^{\sum n_k}$: Scheduled idle wait duration at lower-bounded open windows ($e_i - T_i^k$).
  - $\mathbf{SoC}_{\mathrm{profile}}$: Monotonically decreasing state-of-charge trajectory across route arcs, including recharge events.
- **Output Schema $\mathcal{F}_3$ (Infeasibility Certificate):**
  - $\mathrm{Cut}_{\mathrm{Benders\_Routing}}(\mathcal{V}_{\mathrm{sub}})$: Subtour or temporal impossibility cut returned to Tier 1 indicating that node set $\mathcal{V}_{\mathrm{sub}}$ cannot be serviced by any available vehicle within shift limits H_{shift} or energy capacities.

Tier 4: Kinematic Multi-Agent Execution, Trajectory Deconfliction, and Continuous HRI

- **Input Schema $\mathcal{I}_4$:**
  - $\mathbf{\Sigma}_{\text{routes}}, \mathbf{T}_{\text{arrival}}$: Nominal route sequences and temporal targets from Tier 3.
  - $\mathcal{V}_{\text{ASRS}}$: Scheduled cycle ready-times $T^{\text{ready}}_{\text{crane}}(s)$ for high-bay induction spurs.
  - $\mathcal{Z}_{\text{mixed}}$: Geofenced human-robot co-working zones.
  - $\mathcal{R}^{\text{sweep}}_k(\theta)$: Vehicle rotational swept-envelope geometries.
  - $\mathcal{X}_{\text{static}}$: Metric 2D occupancy grid of structural obstacles (racks, pillars, bollards).
  - $\mathcal{X}_{\text{human}}(t)$: Real-time tracking and state-estimation vectors of personnel inside $\mathcal{Z}_{\text{mixed}}$.
- **Output Schema $\mathcal{O}_4$ (Success):**
  - $\mathbf{p}_k(t) = [x_k(t), y_k(t), \theta_k(t)]^\top \in \mathcal{C}^2$: Continuous, collision-free, C^2 -differentiable kinematic trajectories for $t \in [t_0, t_{\text{end}}]$.
  - $\mathcal{W}_k(t) \subset \mathbb{R}^2 \times \mathbb{R}^+$: Spatiotemporal swept-volume reservation tubes satisfying $\mathcal{W}_k(t) \cap \mathcal{W}_{k'}(t) = \emptyset \quad \forall k \neq k'$.
  - $\mathcal{G}_{\text{priority}}$: Dynamic directed acyclic priority graph governing passage precedence at topological intersections.
  - $\mathbf{U}_{\text{control}}(t) = [v_k(t), \omega_k(t)]^\top$: Low-level actuator control vectors streamed to VDA 5050 / ROS2 controller interfaces.
- **Output Schema $\mathcal{F}_4$ (Infeasibility Certificate):**
  - $\text{Cut}_{\text{Spatiotemporal\_Deadlock}}(k, k', \mathcal{A}_{\text{choke}})$: Spatiotemporal conflict cut proving that physical corridor or intersection $\mathcal{A}_{\text{choke}}$ is kinematically locked (e.g., bilateral head-on encounter on a single-lane segment), forcing Tier 3 to re-route one of the conflicting agents.

Flow Staging and Cross-Tier Lifecycle Orchestration

The optimization pipeline runs on three decoupled, asynchronous execution loops: the

Macro Planning Wave, the **Rolling Route Refinement**, and the **Micro-Second Kinodynamic Loop**.

TIME SCALE PIPELINE EXECUTION ARCHITECTURE

| WMS Dynamic Order Release & Inflow Stream |

Macro-Wave

[10 - 30 min] | TIER 1: Master Batching, Depot Balancing & Chance Alloc |

Event-Triggered

[< 500 ms] | GATE 1: Volumetric & Mass Capacity Bounds Screening |

Batch Synthesis

[1 - 5 s] | TIER 2: 3D Packing, Compression & LIFO Feasibility |

Event-Triggered

[< 100 ms] | GATE 2: Static Equilibrium & LIFO Acyclicity Assertion |

Rolling-Horizon

[1 - 5 min] | TIER 3: Asymmetric Open-Window Routing & Scheduling |

Event-Triggered

[< 50 ms] | GATE 3: Open-Window Slacks & SoC Dynamic Check |

Real-Time Loop

```
[10 - 50 Hz]      | TIER 4: Continuous Swept-SIPP, PBS & Receding D-NMPC  |
                  └───────────────────────────┬───────────────────────────┘
                          |
                          ▼
Sub-millisecond
┌─────────────────────────────────────────────────────────────┐
[Hardware Rate]   | GATE 4: ISO 3691-4 Safety Envelope & Collision Invariance|
                  └───────────────────────────┬───────────────────────────┘
                          |
                          ▼
                  ┌───────────────────────────────────────────────────────┐
                  | Actuator Command Streaming (VDA 5050 / ROS2 Trajectory) |
                  └───────────────────────────────────────────────────────┘
```

2. Cross-Tier Recourse and Benders Cut Propagation

When a downstream tier encounters an irreducible constraint violation, execution halts along that branch and an analytical cut propagates backward through the state machine:

```
┌─────────────────────────────────────┐
|   TIER 1: MASTER BATCHING     |
└──────────────────┬──────────────────┘
      |
 Forward: ℬ_m | ▲ Recourse: Cut_Benders_Packing(ℬ_m^sub)
      | | (Irreducible Infeasible Sub-batch)
      ▼ |
┌─────────────────────────────────────┐
|   TIER 2: 3D CONTAINERIZATION  |
└──────────────────┬──────────────────┘
      |
Forward: 𝒢_LIFO | ▲ Recourse: Cut_Benders_Routing(𝒱_sub)
      | | (Temporal / Subtour / SoC Infeasibility)
      ▼ |
┌─────────────────────────────────────┐
|   TIER 3: ROUTE SEQUENCING    |
└──────────────────┬──────────────────┘
      |
Forward: Σ_rout | ▲ Recourse: Cut_Deadlock(k, k', 𝒜_choke)
      | | (Persistent Kinematic Corridor Lock)
      ▼ |
┌─────────────────────────────────────┐
|   TIER 4: KINEMATIC DECONFLICT. |
└─────────────────────────────────────┘
```

1. Tier 2 to Tier 1 Recourse Protocol (Volumetric / LIFO Violation)

- *Trigger:* Tier 2 CP-SAT or Dual-Chromosome GA determines that sub-batch $\mathcal{B}_m^{\text{sub}} \subseteq \mathcal{B}_m$ cannot be packed without violating static support κ_{support}, compression limits $\sigma_i^{\max}$, or LIFO extraction.
- *Cut Synthesis:* Tier 2 isolates the minimal set of conflicting items $\mathcal{I}_{\text{conflict}} \subseteq \mathcal{B}_m$.
- *Propagation:* Tier 2 returns $\text{Cut}_{\text{Benders\_Packing}}$ to Tier 1:
$$\sum_{i \in \mathcal{I}_{\text{conflict}}} y_{io}^k \leq |\mathcal{I}_{\text{conflict}}| - 1 \quad \forall k \in \mathcal{K}$$
- *Tier 1 Reaction:* Tier 1 updates its master constraint set, immediately splitting $\mathcal{I}_{\text{conflict}}$ across different vehicles or execution waves, and re-solves.

2. Tier 3 to Tier 1 Recourse Protocol (Temporal / Energy Violation)

- *Trigger:* Tier 3 HGS-ADC or Branch-Price-and-Cut discovers that no feasible traversal timing exists for assigned node set $\mathcal{V}(\mathcal{B}_m)$ without violating vehicle shift limits ($T_{d_e}^k - T_{d_s}^k \leq H_{\text{shift}}$) or draining battery reserves below $\text{SoC}_k^{\min}$.
- *Cut Synthesis:* Identification of the critical spatial path $\mathcal{V}_{\text{crit}} \subseteq \mathcal{V}(\mathcal{B}_m)$ whose metric distance forces the temporal violation.
- *Propagation:* Tier 3 transmits $\text{Cut}_{\text{Benders\_Routing}}$ to Tier 1:
$$\sum_{i \in \mathcal{V}_{\text{crit}}} x_{ij}^k \leq |\mathcal{V}_{\text{crit}}| - 1 \quad \forall k \in \mathcal{K}$$
- *Tier 1 Reaction:* The master problem re-clusters $\mathcal{V}_{\text{crit}}$ into an alternate depot catchment area with shorter transit baselines.

3. Tier 4 to Tier 3 Recourse Protocol (Kinematic Deadlock / Infrastructure Fault)

- *Trigger:* Tier 4 PBS encounters a permanent topological confrontation where two vehicles with overlapping swept hulls occupy an aisle segment $\mathcal{A}_{\text{choke}} \in \mathcal{A}_{\text{narrow}}$ with zero safe intervals available ($\mathcal{T}_{\text{safe}} = \emptyset$), or where an AS/RS hoist experiences an operational fault.

- *Cut Synthesis:* Spatiotemporal coordinates of the blockage point $(u, v) \in \mathcal{A}$ and the active time interval $[t_{\text{start}}, t_{\text{end}}]$.
- *Propagation:* Tier 4 injects a time-dependent arc-penalty cut to Tier 3: $c_{uv}^{k}(t) \leftarrow \infty \quad \forall t \in [t_{\text{start}}, t_{\text{end}}]$
- *Tier 3 Reaction:* Tier 3 executes an immediate local ruin-and-recreate operator (ALNS), re-routing the lower-priority agent around the obstructed corridor via alternative transit arcs.

3. Architectural Validation Gates and Invariance Criteria

To prevent corrupt state propagation, strict validation gates sit between consecutive tiers:

Gate Identifier	**Location**	**Validation Operator**	**Invariant Criteria for State Transition**	**Failure Action**
Gate 1	Tier 1 $\rightarrow$ Tier 2	Linear Pre-Screening	$\sum_{i \in \mathcal{B}_m} \mathbf{q}_i \leq \mathbf{Q}_k \quad \wedge$	Reject batch; re-cluster in Tier 1 with reduced cluster radius.
Gate 2	Tier 2 $\rightarrow$ Tier 3	Graph Acyclicity & Conic Equilibrium	$\text{IsAcyclic}(\mathcal{G}_{\text{LIFO}}) \equiv$	Trigger Tier 2 $\rightarrow$ Tier 1 Benders cut; partition conflicting SKUs.
Gate 3	Tier 3 $\rightarrow$ Tier 4	Open-Window Monotonicity & Energy Bounds	$T_i^k \geq e_i \ (\forall i \in \mathcal{V}_{\text{open}}$	Reject route sequence; invoke Tier 3 ALNS battery-recharge insertion.
Gate 4	Tier 4 $\rightarrow$ Hardware	Continuous Minkowski Intersection Test	$\min_t \lVert \mathcal{W}_k(t) \cap \mathcal{W}_{k'}(t$	Emergency hardware halt; issue local receding-horizon D-NMPC evasion.

4. State Synchronization and Shared Execution Memory

To support real-time execution across asynchronous loops without locking overhead, the engine maintains an in-memory **Spatio-Temporal World State**:

```
                 SPATIO-TEMPORAL WORLD STATE
┌──────────────────────────────────────────────────────────────────────────────┐
│ CENTRAL TOPOLOGICAL MAP: Directed Graph 𝒢 = (𝒱, 𝒜) with Dynamic Cost Weights c_ij(t) │
├──────────────────────────────────────────────────────────────────────────────┤
│ DYNAMIC RESERVATION TABLE: Safe-Interval Database 𝒯_safe(v) for all v ∈ 𝒱          │
├──────────────────────────────────────────────────────────────────────────────┤
│ FLEET TELEMETRY BUFFER: 6-DoF Kinematic State Vectors [x, y, θ, v, ω, SoC]         │
├──────────────────────────────────────────────────────────────────────────────┤
│ CONSOLIDATION CHUTE BUFFERS: Real-Time Fluid Queue Heights Q_c(t)                  │
├──────────────────────────────────────────────────────────────────────────────┤
│ AS/RS HOIST REGISTRY: Real-Time Carriage Extraction Timestamps T_crane^ready(s)    │
└──────────────────────────────────────────────────────────────────────────────┘
```

1. **Memory Model:** Double-buffered memory layout with lock-free atomic read snapshots for Tier 3 and Tier 4 route evaluations.
2. **Telemetry Stream Ingestion:** Sub-second sensor feeds from AMR wheel odometry, LiDAR-SLAM poses, and ultra-wideband (UWB) personnel tags continuously update the world state.
3. **Dirty-Bit Invalidation:** If an AMR deviates from its planned trajectory by more than threshold $\delta_{\text{drift}} > 0.3\,\text{m}$ or if a dynamic obstacle blocks a corridor, the corresponding nodes in the Dynamic Reservation Table are flagged as "dirty," immediately triggering a Tier 4 local replanning event.

5. Empirical Falsification and Critical Boundary Analysis

Every rigorous optimization architecture must define the exact conditions under which its descriptive validity breaks down (*verification code: lmn*).

Algorithmic Falsification Protocol

Falsifying Observation: If in an automated fulfillment warehouse running at high order line densities, deploying a **Rank-1 Genetic / Soft-Clustering Pipeline** (FCM $\rightarrow$ 3D CP-SAT $\rightarrow$ HGS-ADC $\rightarrow$ PBS) results in higher total order cycle times and higher frequency of replanning events than a **Rank-3 Static Priority Framework** (Constrained K-Means++ $\rightarrow$ DRL Packing $\rightarrow$ Pulse BPC $\rightarrow$ CCBS), the proposed tiered algorithmic hierarchy is falsified.

- **Failure Mechanism:** Soft-clustering (FCM) and genetic metaheuristics maximize solution diversity by exploiting wide solution neighborhoods. However, soft-clustering generates overlapping vehicle paths that intersect repeatedly across common corridors.
- **Physical Gridlock:** Under high facility utilization ($> 85\%$), the computational cost of resolving spatiotemporal conflicts in Tier 4 (PBS) grows non-linearly with route overlap. If the upstream "better" metaheuristic routes from Tiers 1 and 3 create aisle congestion that overwhelms low-level deconfliction, the decoupled tiered hierarchy suffers from downstream infeasibility cascades.
- In this regime, simple orthogonal partitioning (K-Means++ with strict non-overlapping aisle corridors) completely eliminates Tier 4 conflict-checking overhead, rendering the simpler model empirically superior in physical throughput.